The Anatomy of a Truth Direction

Knowledge-Dependent Dimensionality, a Relational Law,
and a Convergent Category Geometry in Small Language Models

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Code: `github.com/Francesco-Marhel/TruthProbe` (one command per reported number)

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Abstract

Bürger et al. (2024) demonstrated that truth representations in large language models are universal across statement polarity but reside within a multidimensional subspace. We extend that framework along three questions: how the dimensionality of the subspace depends on the model's knowledge, which architectural component builds the truth direction, and what the direction is a mixture of. In Part I (one model), a training-free directional probe derived from the SVD of hidden-state minimal pairs shows that the dimensionality of truth is knowledge-dependent: the signal is concentrated on a single axis for behaviorally known facts (held-out AUC 0.938) and becomes diffuse as knowledge decreases or material heterogeneity increases; seven falsification experiments sustain the one-dimensional reading, and the same decomposition recovers the supervised polarity direction of Bürger et al. at cosine 0.959 without polarity labels. In Part II (four base models, two families, pre-registered predictions), a relational law emerges: attention propagates truth frames it did not write, the feed-forward network opposes the frame of its current block, and the genuine post-peak decay is causally attributed on all four models to the SwiGLU value stream; per-category truth axes form a semantically signed arrangement that contradicts the lexical surface and converges across families (Mantel $p = 0.0009$). Stress tests at three scales (33 categories, 888 pairs per relation, six sampling seeds) expose a sign instability of the per-category orientation, repair it with a declared spectral consensus gauge, and sharpen the convergence into a knowledge-gated law, strengthening from +0.42 on dense mixed-knowledge material to +0.74 on knowledge-restricted categories: the geometry of the mixture belongs to the knowledge domain rather than to the architecture.

1 Introduction

Large language models produce false statements with the same fluency as true ones, and a growing body of work shows that the truth value of a statement is linearly readable from the residual stream [1, 2, 3, 4]. Bürger, Hamprecht and Nadler [5] sharpened this picture in a way that motivates everything below: truth is *universal* across statement polarity, but it is not a single direction; probes trained on affirmative statements fail on negated ones, and the representation lives in a

*This manuscript consolidates the author's series on the truth direction: the single-axis paper and the anatomy note (Part I, reviewed in an earlier form and revised following referee-style comments by B. Nadler) and three working notes on the cross-family mechanics and category geometry (Part II). Every number in both parts was regenerated under a single uniform protocol on July 9–10, 2026; the stress tests of Part II were run July 15–17, 2026 under the same contract. Corrections of the author's own published interpretations are reported inside the text, as results.

two-dimensional subspace with a dedicated polarity direction. If the truth representation has more structure than one direction, three questions follow naturally. How much of it fits on a single direction, and when does that reading fail? Which architectural component builds the direction, and which degrades it? And what is the direction made of, when the material spans many kinds of facts?

This manuscript answers the three questions in two parts, with one instrument and one discipline. The instrument is a training-free axis: the dominant direction of the SVD of hidden-state differences over true/false minimal pairs, identified without labels up to one global sign (Lemma 1), read at cost $O(d)$ per token, and evaluated held-out with a permutation null that prices the optimism of layer selection. The discipline is falsification: predictions are registered before runs, negative results are reported with the same care as positive ones, and two corrections of our own published interpretations are narrated in the text as results, because a claim that has survived its author's own best attack is worth more than one that has never been attacked.

Part I (one model, Qwen2.5-1.5B) establishes the object. The axis carries a real signal on facts the model knows, in the behavioral sense defined in Sec. 4.2 (held-out AUC 0.938 against a full-probe ceiling of 0.989); the gap between axis and probe grows as the model's behaviorally measured knowledge decreases, and grows further when heterogeneous material forces one axis to serve many kinds of truth: dimensionality is *knowledge-dependent*. Seven falsification experiments designed to expose a richer structure (a second dimension, depth-global signatures, dynamics, departments, energy, phase, three-dimensional clustering) fail to overturn the one-dimensional reading on known facts. Blind to polarity labels, the same decomposition recovers the supervised polarity direction t_P of [5] at cosine 0.959. An exact attention/FFN decomposition then locates the axis anatomically: attention builds the signal in the middle layers, causally, and the FFN degrades it after the readability peak.

Part II (four base models, two families) turns the anatomy into laws. The FFN's apparent pro-truth write at the peak and flip to anti-truth at the next block, replicated under pre-registered predictions, is exposed by an axis-provenance control as the special case of a *relational law*: the FFN opposes every truth frame that does not contain its own write, while attention propagates frames it did not write (Empirical Law 1); at the peak the two contributions are near mirror images. The genuine post-peak erosion survives the correction and is causally attributed, on all four models and with predictions registered for the second family, to the SwiGLU value stream rather than the gate. Two further pre-registered discriminations return opposite verdicts, showing that scale and training recipe leave distinct signatures: the post-peak attention–FFN tug-of-war is a trait of scale, single-layer ignition of the signal a trait of one family. A pre-registered negative extends cross-family: the gauge-invariant OV/QK circuit crossover sits at a stable late relative depth on all four models and does not track the truth peak. Finally, the axis is opened up: per-category truth axes form a semantically signed arrangement that reorganizes with depth, contradicts the shared lexical arrangement both families start from, and converges across the families; the FFN writes truth along category-dependent directions, decoding the category far above a lexical baseline. Three stress tests then push the geometry to a larger category count, a higher pair density, and re-sampled pairs; they expose, and repair with a declared spectral gauge, a sign instability of the per-category orientation, and they sharpen the arrangement claim into a knowledge-gated law. The knowledge-dependent gap of Part I is, in hindsight, this mixture seen from one direction.

The two parts share every methodological choice: the same pinned dataset revision, the same held-out protocol over pairs, the same permutation-null estimator, the same behavioral know-rate measured before any geometry, and the same reproducibility contract, one command per reported number.

2 Related work

Linear truth directions. [1] extract a knowledge direction without labels by enforcing logical consistency across a statement and its negation; the direction is found by optimization. [2] show truth has emergent linear structure and that difference-of-means probes generalize as well as more complex ones. [3] intervene along truthful directions at inference time; [4] frame representation reading top-down. Our axis shares the linear-direction hypothesis, but it is read from the geometry of paired differences by an SVD, with no optimization; its precise relation to supervision is stated in Lemma 1.

A two-dimensional truth subspace. Closest to us, Bürger et al. [5] show that truth is represented not in a single direction but in a two-dimensional subspace spanned by a general truth direction t_G and a polarity-sensitive direction $t_P \sim \mathrm{XOR}(\mathrm{is\_true}, \mathrm{is\_negated})$, and that probes trained on affirmative statements fail on negated ones. We confirm this structure in Qwen2.5-1.5B and show, as a methodological addition, that t_P can be recovered *without* polarity supervision.

Cross-layer dynamics. Recent monitoring methods model the evolution of hidden states across depth rather than reading a single layer [13, 14]. Our depth trajectories are descriptive and consistent with this line.

Beyond single-direction probes on the residual stream. All the methods above, ours included, assume white-box access to the residual stream, which is not always available and is not the only usable signal. Bar-Shalom et al. [10] treat the full activation tensor (layers by tokens) as an image and train a vision-transformer detector on it, across several LLMs simultaneously, with evaluation on a broad set of hallucination benchmarks. Methods of this kind are more expressive than any single direction and are trained for detection; our axis sits at the opposite end of the same cost/expressivity curve, as the cheapest structural point, and the comparison we draw is against that curve, not against any single method.

Superposition and dictionaries. Superposition [8] motivates decomposing the residual stream into overcomplete dictionaries [6, 7]. This is the operation we explicitly do *not* perform. The bottom line, argued in Secs. 8 and 10, is that the two approaches answer different questions and neither subsumes the other: a dictionary answers "which features are active" at cost $O(d\,m)$ and after dedicated training; the axis answers "how much truth a single direction carries" at cost $O(d)$ with no training. When feature identity matters, dictionaries are the right tool; for a cheap runtime signal on facts the model knows, the axis suffices. We make no claim of superior accuracy over dictionary methods.

MLPs as key-value writers. The memory-editing literature [12] attributes to transformer MLPs the physiology of key-value stores whose writes deposit factual content into the residual stream. The relational law of Sec. 17.3 is a dynamical face of that picture: the component that deposits the next frame's material is, per these measurements, the one that opposes the current frame.

Rotational dynamics of factual selection. A recent preprint [15] reports that factual constraint processing separates alternatives by rotation, with active suppression of the correct continuation at intermediate depth. The propagate/overwrite asymmetry supplies a candidate microscopic writer for that macroscopic observation.

Attention-head specialization. The head-taxonomy line of work [16] motivated the pre-registered test of whether static, gauge-invariant circuit norms predict where truth lives (they do not, Sec. 20) and defines the per-head follow-up of Sec. 24.

Cross-model representation convergence. The convergence of the category arrangement across families (Sec. 21) is a signed, truth-specific instance of the broader phenomenon that different models trained on the same world converge to similar representations [17].

Part I: The Axis on One Model

3 Scope and evidence

Everything in this paper concerns one small base model, Qwen2.5-1.5B, and sentence-level factual statements. Results come in two evidential tiers, and we keep them distinct throughout. **Tier 1** (Secs. 5 and 6): the positive findings and the falsification battery, evaluated held-out with permutation nulls. **Tier 2** (Sec. 7): the anatomy, a mechanistic follow-up with the same protocol but smaller margins in places, single datasets for some entries, and no falsification battery of its own; where a statement there is interpretation rather than measurement, it is marked as such. Cross-model generalization is not claimed here; what is known about it from subsequent work is summarized, with numbers, in Part II.

4 Method

4.1 Extraction

Informally: a transformer reads a sentence and, at every layer, updates an internal vector summary of the text so far; we record that summary at the position of the last token, where the sentence is complete, at every layer, and study the geometry of these vectors.

Pooling and the token-identity confound. All states are read at the last token, and in minimal pairs the flipped content word sits at or immediately before that position: token identity is therefore a candidate confound for everything downstream. Three measurements bound it. First, the per-layer curves start at chance in every scan (layer-0 AUC = 0.500 on all four models): a token-identity axis would separate already at the embedding, and it does not; the signal emerges only with depth, that is, with context integration. Second, all AUCs are held-out across pairs whose targets are disjoint, which a pure target-identity direction would not survive at the measured levels. Third, the category analysis carries its own lexical control: the block-2 baseline decodes the relation at 39–42% against 74–82% for the peak-adjacent FFN writes. The extraction tool also supports mean pooling (`-pool mean`); the last-token results are the canonical ones, and the mean-pooling replication is listed as a cheap robustness run rather than assumed.

Formally, we load Qwen2.5-1.5B frozen. The model has 28 transformer blocks, hence 29 hidden-state levels (the embedding plus each block output), each a vector of width $d = 1536$, the width of the residual stream. Per sentence we run one forward pass and take the last-token state at every level. For N sentences this yields $H \in \mathbb{R}^{N\times 29\times d}$. Once an axis is fixed, scoring a new state is a single dot product, hence the reading cost of $O(d)$ per token quoted throughout.

4.2 Data: minimal pairs

A *minimal pair* is a pair of sentences with the same subject and the same template that differ in exactly one content word, such that one sentence is true and the other false: "The Sun is a star." vs. "The Sun is a planet." Both sentences are affirmative. Within a pair, the difference of hidden states cancels the shared topic and surface form and leaves the truth signal.

Two sources are used. The *curated pairs* are a built-in set of 45 pairs of common facts (capitals, chemical symbols, authorship, currencies) that the model demonstrably reproduces in open-ended completion (Sec. 11); they play the role of a known-facts control: behaviorally verified and token-matched, hence also clean, in the sense of free of the sentence-format confounds discussed in Sec. 12. The external datasets are CounterFact [11] and TruthfulQA [18] (250 pairs, pinned revisions in Sec. 27), which contain many relations a 1.5B model plausibly does not know.

To quantify this, we define the *know-rate*: the fraction of facts the model reproduces correctly in open-ended greedy completion. Over 200 CounterFact prompts, Qwen2.5-1.5B has a know-rate of 30.0% (61/200), which grounds the "plausibly does not know" reading; the know-rates of the other models appear in Part II.

Precisely: a prompt counts as known when the greedy completion (`do_sample=False`, at most 8 new tokens) either begins with the true target or contains it preceded by a space, case-insensitively (`behav_check.py`). Three properties of this estimator matter for what follows. It is a *lower bound*: a model may know a fact and still continue differently within eight tokens. It is *greedy*: no sampling, so the estimate is deterministic given the seed that selects the prompts. And it inherits tokenizer and completion-style differences between families, so cross-family comparisons carry a systematic bias component, while within a family the bias is shared and cancels. This is why the dimensionality relation is stated as within-family monotone with a cross-family offset, and why the ordering Llama-3.2-1B (35.5%) above Qwen2.5-1.5B (30.0%) is neither exotic nor load-bearing: parameter count is not knowledge (the two models differ in training data, tokenizer, and completion style), and no claim in this manuscript rests on comparing know-rates across families.

Why a flipped predicate and not a negation. The false counterpart is built by flipping a content word ("planet" for "star"), not by inserting "not". The choice is deliberate, for three reasons. First, negation is not a neutral edit: it activates a dedicated polarity direction in the representation [5], so a negated counterpart would entangle the truth signal with the polarity signal. Our design keeps both sentences affirmative and isolates truth; polarity is then studied separately and deliberately in the 2×2 design of Sec. 4.6. Second, inserting "not" changes the token count and shifts all subsequent positions, weakening the token-level matching that makes the intra-pair difference cancel shared structure. Third, the two designs are complementary rather than competing: the negated forms are exactly the NT/NF cells of the polarity design, so the alternative is not ignored but treated as its own object of study.

Lexical confounds, declared up front. Any paired design risks measuring the lexical surface rather than truth: the flipped content word, the template's wording, or both. Four controls, executed in later sections but declared here because a careful reader is right to demand them early, bound the risk. The per-layer curves start at chance (layer-0 AUC 0.500 on all four models of Part II), which a token-identity axis would not do; all AUCs are held out across pairs with disjoint targets; the category analysis carries its own lexical baseline (decoding the relation from early residual differences reaches 39–42% against 74–82% for the peak-adjacent FFN writes); and the early category arrangement is uncorrelated or anti-correlated with the peak arrangement within each model ($r = -0.547$ and -0.175), so the peak geometry contradicts the lexical surface it started from. One residual is declared, not dismissed: each CounterFact relation contributes one statement template, so template syntax can contribute to individual cells (the continent anti-alignment is the flagged case), and crossed templates are the known cure.

4.3 The axis, and where labels enter

We form the intra-pair difference matrix D with rows $h_{\text{true}} - h_{\text{false}}$ at a given layer, and take its SVD,

$$D = USV^\top, \qquad v_1 = V_1, \tag{1}$$

where v_1, the top right-singular vector, is the dominant direction of the differences. The role of supervision in this construction can be stated exactly.

Lemma 1 (Label-free identification of the axis, up to sign)**.** *Let each pair i contribute a row $d_i^\top$ to D, where $d_i = s_i\,(h_i^A - h_i^B) \in \mathbb{R}^d$ is a column vector, h_i^A, h_i^B are the two hidden states of the pair, and $s_i \in \{\pm 1\}$ encodes which sentence of the pair is designated true. Then $D^\top D = \sum_i d_i d_i^\top$ does not depend on (s_i); hence the right singular vectors of D, and in particular the span of v_1, are invariant to the within-pair labels.*

One implicit choice is declared here rather than discovered by a referee: the SVD is applied to the raw difference rows, so v_1 is a *norm-weighted* principal direction, with pair i contributing weight proportional to $\|d_i\|^2$. If difference norms correlate with sentence length or with the state norm at the reading position, long or high-norm pairs dominate the fit. The control was executed on all four models at their peaks (`axis_norm_check.py`): the row-normalized refit agrees with the raw axis at $|\cos| = 0.984$, 0.980, 0.974, 0.966 (Qwen2.5-1.5B, Qwen2.5-3B, Llama-3.2-1B, Llama-3.2-3B), the held-out AUC changes by at most +0.016, and is in fact slightly higher under normalization on every model, so the axis does not ride on high-norm pairs; difference norms correlate only weakly, and negatively, with sentence length (r between -0.12 and -0.23). The axis is not a norm-weighting artifact.

Proof. $d_i d_i^\top = (-d_i)(-d_i)^\top$ for every i, so each summand of the Gram matrix, and therefore the Gram matrix and its eigenvectors, is unchanged under any sign assignment. □

What label-free does and does not mean. Three levels of supervision must be kept apart. (i) *Dataset construction* is supervised: writing a minimal pair requires knowing which completion is factually correct, and no method that starts from such pairs can call itself unsupervised at this level. (ii) *Axis estimation* consumes only the contrast structure: by Lemma 1, the span of v_1 is invariant to which element of each pair is designated true, so an adversary who scrambled every within-pair label would hand the algorithm the same axis. The information the estimator uses is "these two statements differ in truth value", not "this one is true". (iii) *Orientation and evaluation* are supervised: one global sign bit, and held-out labels for scoring. Accordingly, this manuscript says *training-free* of the pipeline, *label-free up to sign, given the paired design* of the estimator, and reserves *unsupervised* for the polarity recovery of Part I, where the recovered factor was never encoded in any label handed to the algorithm.

Labels enter the pipeline at exactly two points. First, a single global orientation bit ("the true side is positive"), fit on training folds, fixes the sign of v_1. Second, the evaluation itself: AUC is defined with respect to labels. Everything upstream of these two points, the pairing, the differences, and the axis, uses no labels. We therefore describe the axis as *training-free* (no optimization of any kind) and *label-free up to one sign bit* (Lemma 1), and we reserve the word "unsupervised" for the polarity recovery of Sec. 5.4, where the recovered structure (polarity) is never labeled at all. The lemma also explains, in advance, the behavior of the permutation null below: permuting which sentence of a pair is called true leaves the span of the axis unchanged and destroys only the orientation and the test labels, which is why the null concentrates near chance.

4.4 A falsifiable evaluation protocol

The central methodological point is that the axis, its calibration, the orientation bit, *and* the choice of peak layer (the layer whose held-out AUC is maximal, chosen on training folds) must be fit on training data only. We use k-fold cross-validation *over pairs*: a pair is never split across folds, which would leak the shared topic. Everything is fit on the training folds and we report held-out AUC averaged across folds. Pairs from the same CounterFact relation template may still fall in different folds; generalization across topics is therefore not certified by this protocol, and we return to the point in Sec. 12.

Part of the apparent accuracy of any layer-scanned pipeline is selection: the best of 29 layers looks good by chance. The *permutation null* quantifies exactly that contribution. Let A_{obs} be the observed maximum held-out AUC over layers (AUC: the area under the ROC curve; equivalently, the probability that a randomly chosen true statement scores above a randomly chosen false one, ties counted as one half). For $b = 1, \dots, B$ we randomly swap true/false *within* each pair (the labels become noise while the pair structure is preserved), rerun the identical pipeline including the layer scan, and record the maximum held-out AUC A_b. The estimated p-value is the standard conservative estimator

$$\hat{p} \;=\; \frac{1 + \#\{\, b : A_b \geq A_{\text{obs}} \,\}}{B + 1}. \tag{2}$$

With $B = 200$ and no null value reaching the observed statistic, $\hat{p} = 1/201 \approx 0.005$, the value reported in Table 1. As a reference point we also fit an ℓ_2-regularized logistic probe on the raw d-dimensional states, held out over the same pair folds.

4.5 A complex-plane extension

We additionally form a second axis v_2, the SVD residual orthogonal to v_1, treat the projection as $z = c_1 + i\, c_2$ with calibrated real and imaginary parts, and test whether magnitude- or phase-based scores improve separation over the real axis alone. They do not (Sec. 5.3); the negative finding is kept because it is itself informative.

4.6 Polarity design

To test whether a genuine second dimension exists, we use a 2×2 design: each fact in four forms, affirmative-true (AT), affirmative-false (AF), negated-true (NT), negated-false (NF), giving the XOR structure of [5]. We compute the affirmative and negated truth directions $t_A = \overline{AT} - \overline{AF}$, $t_N = \overline{NT} - \overline{NF}$, the general direction $t_G \propto t_A + t_N$, and the supervised polarity direction $t_P \propto t_A - t_N$. For the recovery test, we run the difference SVD on minimal pairs that *mix* the two polarities, so the SVD never sees the polarity label, and compare its leading directions to the supervised t_P.

4.7 Residual decomposition and its correctness gate

Within a block the residual stream is updated additively,

$$h_{\ell+1} \;=\; h_\ell + a_\ell + f_\ell, \tag{3}$$

Table 1: Held-out AUC (5-fold CV over pairs) at layer 16, Qwen2.5-1.5B. "1D" is the real axis; "MAG" the magnitude-gated polar score; "PHASE" the angular-deviation score; "2D" a logistic on $(\mathrm{Re}, \mathrm{Im})$; "Probe" an ℓ_2 logistic on raw states. The last column is the label-permutation $\hat{p}$ for the 1D axis.

Dataset	pairs	1D	MAG	PHASE	2D	Probe	$\hat{p}$
Curated pairs	45	0.938	0.936	0.928	0.931	0.989	0.005
CounterFact only	250	0.792	0.790	0.771	0.795	0.854	0.005
CounterFact / TruthfulQA mix	250	0.638	0.640	0.676	0.651	0.841	0.005

where a_ℓ is the attention contribution and f_ℓ the FFN contribution, both captured by forward hooks. The residual stream studied everywhere else in this paper is the sum; the anatomy of Sec. 7 reads the two addends separately.

Every anatomy tool embeds an **identity check**: it verifies that $a_\ell + f_\ell$ reconstructs the residual delta $h_{\ell+1} - h_\ell$. The verdict uses the *median* relative error across (sentence, layer) rather than the maximum: where a block adds almost nothing the delta is near zero and the ratio explodes, so the maximum is fragile while the median reports the typical case. Observed median error: about 1.0×10^{-7} on both datasets, so the decomposition is exact and the anatomy numbers are trustworthy. (The single inflated layer, block 27, has a near-zero residual delta and is excluded from interpretation.) Anatomy runs use `float32`: the identity fails spuriously in `bfloat16` because subtracting large residual states to form small per-block deltas suffers catastrophic cancellation.

5 Results I: signal, dimensionality, polarity

All numbers are on Qwen2.5-1.5B with last-token pooling; AUC is out-of-fold unless stated.

5.1 A real but bounded truth signal

Table 1 reports the held-out evaluation at the peak layer (layer 16). On the curated pairs the training-free axis reaches AUC 0.938 with permutation $\hat{p} = 0.005$ (null mean 0.588, 95th percentile 0.664): the geometric axis aligns with truth well beyond what layer selection alone can manufacture. The signal is therefore real. It is also *bounded*: a full linear probe on the same raw states reaches 0.989. The probe sees information the single axis discards.

5.2 The gap scales with model knowledge, and with heterogeneity

The rows of Table 1 differ not only in difficulty but in the *size of the probe gap*, and the three rows separate two distinct effects.

The first effect is knowledge. On curated facts the model demonstrably knows, the axis (0.938) trails the full probe (0.989) by 0.051. On CounterFact alone, where the measured know-rate is 30.0% (Sec. 4.2) and many relations are obscure (for example, the mother tongue of a minor historical figure), the separability of the axis falls to 0.792 and the gap widens to 0.062 (probe 0.854). Less knowledge, lower separability, and a modestly larger gap.

The second effect is heterogeneity. On the CounterFact/TruthfulQA mix the gap jumps to 0.203 (axis 0.638, probe 0.841), far more than either ingredient alone would suggest. A single direction fit across heterogeneous material must serve all of it at once, while the full probe is free to combine directions; the excess gap on the mix therefore measures how much the truth direction itself varies with the material. Subsequent work confirms this reading independently Part II: the per-category truth axes form a structured, semantically signed geometry, and forcing one axis onto a mixture of categories is exactly what the mix row pays for.

We read the knowledge effect as a statement about *dimensionality*. When the model knows a fact, its truth content is concentrated and a single direction captures most of it. When the model does not know a fact, the truth-relevant signal is weaker and more diffuse, and one direction recovers less of it. The dimensionality of the truth representation is thus not fixed but *knowledge-dependent*; the mix row adds the complementary statement that it is also *category-dependent*. This joint framing is, to our knowledge, stated in these terms by neither the linear-probe nor the dictionary-learning literature, and it is the contribution we consider most novel.

5.3 The complex plane does not help; "phase" is the sign flip

On single-polarity data the second dimension adds nothing: at layer 16 the 2D logistic (0.931) does not exceed the 1D axis (0.938) on curated pairs, and on CounterFact it gains +0.003 (0.795 vs. 0.792). This is structural. Minimal pairs cancel the topic, so the differences concentrate along one direction; the SVD places the entire signal in v_1, and v_2 is residual variance, noise rather than a second truth factor. A controlled synthetic check confirms that the apparent "rotation" of falsehoods is an artifact: with a pure sign flip and zero genuine rotation, an angular (phase) score still separates classes, because a state at $\mathrm{Re} < 0$, $\mathrm{Im} \approx 0$ has phase $\approx \pi$ by definition. The earlier intuition of truth-as-phase has no operational content here: it reduces to the sign of the real axis re-expressed in polar coordinates. We retract that framing.

5.4 Recovery of the polarity direction without polarity labels

A genuine second dimension does exist, but only when the data span both polarities, and it is the polarity direction of [5], not a phase. Table 2 shows the diagnostic: an axis fit on affirmative statements classifies affirmative held-out facts at 0.981 but *anti-classifies* negated ones at 0.079, a near-total flip, while the general direction t_G recovers the negated facts at 0.954. The truth direction genuinely rotates with polarity ($\cos(t_A, t_N) = -0.415$).

The methodological result is the recovery. When we run the difference SVD on *mixed-polarity* minimal pairs, blind to the negation label, its leading direction aligns with the supervised polarity direction t_P at cosine 0.959 (layer 15), and t_P lands in the first principal component, while alignment with t_G is only 0.157. In other words, when both polarities are present, the dominant axis of the differences *is* polarity, and the SVD finds it without supervision; by Lemma 1 the recovery cannot have used the labels even implicitly. Where [5] constructs t_P by separating affirmative from negated statements by hand, the same direction emerges here from geometry alone.

6 Robustness: seven falsification experiments

The first half of this paper established a positive claim. We now stress-test it. The claim under test is:

> *In Qwen2.5-1.5B, the truth value of a statement is carried by a single (one-dimensional) direction, the position axis obtained from the SVD of paired true/false differences, on facts the model knows; the second and higher dimensions add no separation.*

Table 2: Polarity analysis (leave-facts-out CV), layer 14 chosen on affirmative data; and unsupervised t_P recovery at layer 15. AUC below 0.5 indicates a sign flip.

Quantity	Value
$\cos(t_A, t_N)$	-0.415
Affirmative axis on affirmative (held-out AUC)	0.981
Affirmative axis on *negated* (held-out AUC)	0.079
General direction t_G on negated (held-out AUC)	0.954
SVD vs. supervised t_P: $\max_k \lvert\cos(\mathrm{SVD}_k, t_P)\rvert$	0.959
(principal component carrying t_P)	PC 1
$\cos(\mathrm{SVD}_1, t_G)$	0.157

A claim of one-dimensionality is falsifiable in a specific way: exhibit a second, third, or depth-distributed coordinate that separates true from false better than the single axis. We construct seven such coordinates and test each.

Two kinds of test, kept distinct. We separate *controlled experiments* from *visual observations*, and label every result as one or the other. Controlled experiments (experiments 1–4) produce a held-out AUC under k-fold cross-validation over pairs and a label-permutation null that quantifies selection bias; a result is meaningful only if it clears that null. Visual observations (experiments 5–7) do not classify; they project the states and measure a single descriptive statistic, the per-coordinate separation AUC or the centroid-separation ratio, so the conclusion follows from a number attached to a picture, not from a fitted model. Throughout, the model is frozen Qwen2.5-1.5B, last-token pooling, peak layer 16 unless stated.

Table 3 summarizes all seven with their real numbers; the prose below gives the reasoning for each.

1. The complex plane. The original framing promoted the second SVD direction to an imaginary axis and read truth as a phase rotation. If real, a phase- or magnitude-based score should beat the real axis. It does not: on curated pairs the phase score reaches AUC 0.928 against 0.938 for the real axis, and a controlled synthetic check shows that with a pure sign flip and *zero* genuine rotation an angular score still separates, because a state at $\mathrm{Re} < 0$ has phase $\approx \pi$ by definition. The "rotation" is the sign of the real axis re-expressed in polar coordinates. The complex framing is decorative; we retract it. (This is the negative result of Sec. 5.3, restated as the first falsification experiment.)

2. A global depth signature. If truth flows through the network, the full trajectory $[\Delta_0, \ldots, \Delta_{28}]$ should beat the single best layer, where the signed deviation at each layer ℓ is defined as $\Delta_\ell = b_\ell - \frac{1}{2}$, with $b_\ell = \sigma(\mathrm{Re}(z_\ell))$ representing the calibrated projection passed through a sigmoid along a fixed axis. On curated pairs it does not: the depth-signature classifier reaches held-out AUC 0.923 against 0.938 for the single layer (-0.015), and although the signature is real (permutation $\hat{p} = 0.0099$), it carries no advantage. The crossing-persistence gap between false and true is $+0.04$, flat (where for each sentence, the first layer at which the trajectory crosses below zero and the fraction of subsequent layers it stays below; the gap is the difference of this persistence between false and true statements). No propagation.

3. A dynamic butterfly effect in the shade. The signed deviation saturates (σ compresses large values to $\pm\frac{1}{2}$), so a propagating *uncertainty* could be invisible to it. We re-ran on hard

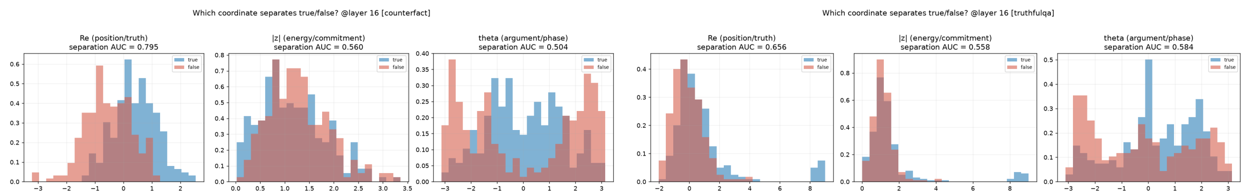
Table 3: Seven experiments attempting to falsify the one-dimensional truth axis. "Exp." = controlled experiment (held-out CV + permutation null); "Obs." = visual observation (descriptive separation statistic). None overturns the claim.

#	Richer structure hypothesized	Type	Result (real numbers)	Verdict
1	A complex/phase second dimension (truth as rotation)	Exp.	PHASE AUC 0.928 vs. 1D 0.938 (curated); reduces to the real-axis sign flip	falsified
2	Truth is global across depth (fixed-axis signature)	Exp.	global 0.923 vs. local 0.938 (-0.015), $\hat{p} = 0.0099$; persistence gap +0.04	falsified
3	Truth is dynamic, a "butterfly effect" in the unsaturated shade, on hard data	Exp.	global 0.642 vs. local 0.627 (+0.015), $\hat{p} = 0.0099$; commitment true $2.83 \to 5.91$ vs. false $2.65 \to 5.86$	falsified
4	Truth is distributed across layer "departments"	Exp.	diagnostic curve is a smooth ramp (no steps); global 0.646 vs. local 0.627 (+0.019), $\hat{p} = 0.0099$	falsified
5	The argument θ (phase) carries truth	Obs.	θ AUC 0.504 / 0.584 / 0.556 (CF / TQA / mix)	falsified
6	The energy $\lvert z\rvert$ (commitment) carries truth	Obs.	$\lvert z\rvert$ AUC 0.560 / 0.558 / 0.543	falsified
7	True/false form distinct 3D clumps	Obs.	centroid ratio 0.55 / 0.51 / 0.48 (< 1, overlapping); Re alone 0.795 / 0.656	falsified

data (CounterFact/TruthfulQA mix) with the unsaturated features $[\mathrm{Re}, |z|]$, the "shade". The depth signature now beats the single layer, but by +0.015 (0.642 vs. 0.627, $\hat{p} = 0.0099$): real and marginal. The decisive descriptive measure kills the butterfly effect: the commitment magnitude grows identically for both classes (true $2.83 \to 5.91$, false $2.65 \to 5.86$); false statements do not hesitate more than true ones, and there is no propagating divergence.

4. Layer departments. If different layer bands perform different jobs (surface, semantics, logic), a single semantic axis re-read everywhere wastes most layers; giving each layer its *own* native probe should recover a distributed signal, and the per-layer separation curve should show steps. It shows a smooth ramp instead: the per-layer native separation rises continuously to a single peak at layer 16, with no plateaus and no jumps. The per-layer signature beats the single layer by +0.019 (0.646 vs. 0.627, $\hat{p} = 0.0099$), again real and marginal. No departments; the native-axis "cure" improves the fixed axis but does not change the ceiling.

5–6. Energy and argument. Decomposing each state into position (Re), energy ($|z|$), and argument (θ) and measuring how well each separates true from false (Fig. 1): only position carries signal. The energy separates at $|z|$ AUC 0.560/0.558/0.543 (CounterFact/TruthfulQA/mix): true and false are equally committed. The argument separates at θ AUC 0.504/0.584/0.556, essentially chance; what little it shows is the echo of the Re sign, not independent information. The phase, the very coordinate the complex framing made central, is empty.



(a) CounterFact: Re 0.795, $|z|$ 0.560, θ 0.504 (b) TruthfulQA: Re 0.656, $|z|$ 0.558, θ 0.584

Figure 1: Experiments 5–6 and knowledge scaling, layer 16. Per-coordinate separation AUC. On both datasets only the position axis (Re) separates true from false; energy ($|z|$) and argument (θ) are at chance. The position separation falls from 0.795 (CounterFact, often-known facts) to 0.656 (TruthfulQA, more ambiguous), while the one-dimensional structure is unchanged.

7. Three-dimensional clumping. If truth were a richer object, true and false should form distinct clusters in the three-dimensional (v_1, v_2, v_3) space. They do not (Fig. 2). The centroid-separation ratio, the distance between the true and false centroids divided by the average cloud spread, is 0.55/0.51/0.48, all below 1: the clouds overlap. Crucially this low ratio is *not* evidence against the axis; it is the one-dimensional signal seen through two noise dimensions. Projected onto position alone, the same data separate at AUC 0.795 (CounterFact). Adding the orthogonal dimensions dilutes, it does not enrich.

6.1 What survives

Three statements resist all seven experiments.

Truth is one-dimensional and lives on the position axis. Across every decomposition, only Re separates; energy, phase, the second and third SVD directions, and the full-depth trajectory add at most +0.02 AUC and usually nothing.

Separability scales with knowledge. On CounterFact, where the model often knows the fact, the position axis separates at AUC 0.795; on the more ambiguous TruthfulQA at 0.656; on the diffuse mix the centroids nearly coincide. The clearer the model's knowledge, the more one-dimensional and separable the truth representation.

On unknown facts, the classes show low separability. Where the model does not know the fact, true and false representations do not form distinct clusters but overlap in space, intuitively a dense, uniform distribution rather than two distinct clumps. We state this as *low class separability / distribution overlap* (centroid ratio < 1), the measured counterpart of the knowledge-dependent dimensionality of Sec. 5.2.

7 Results II: the anatomy of the axis

The falsification battery established *that* truth is one-dimensional on known facts. This section asks *how the residual stream comes to carry it*: which architectural component, attention or the FFN, produces, holds, or degrades the signal, layer by layer. The results here are Tier 2 in the sense of Sec. 3: same protocol and safeguards, but smaller margins in places, and interpretations are marked as such. The decomposition and its correctness gate are defined in Sec. 4.7.

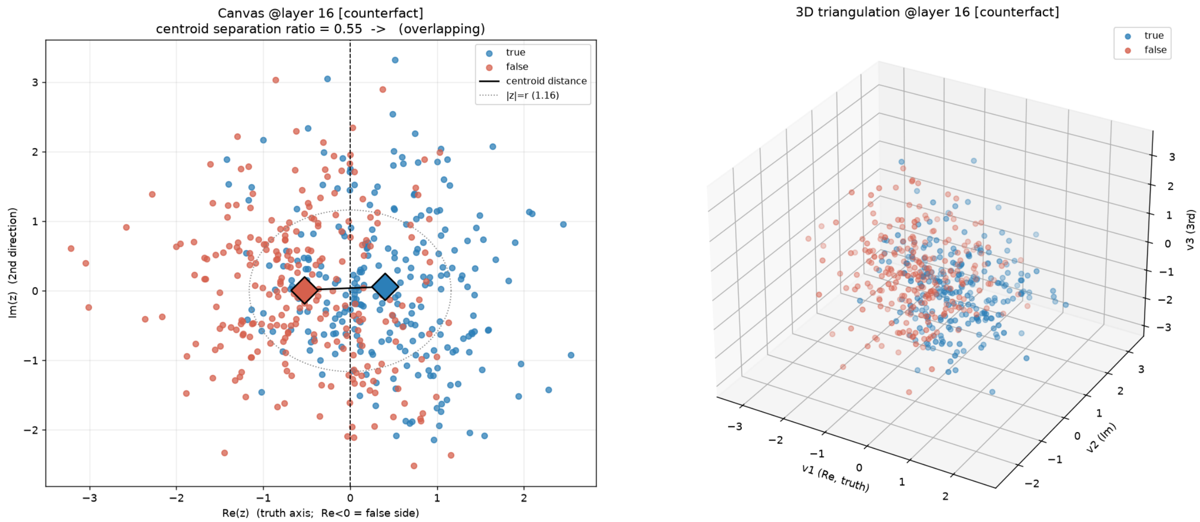


Figure 2: Experiment 7, CounterFact at layer 16. Left: the 2D plane; the true (blue) and false (red) centroids (diamonds) are close relative to the cloud spread (centroid-separation ratio 0.55, overlapping). Right: the three-dimensional (v_1, v_2, v_3) scatter shows a gradient along v_1 (position) but no distinct clumps. The separation is confined to the horizontal position axis; the extra dimensions are noise.

7.1 Per-layer readability: a stable alternation

For each layer we fit the truth axis on a_ℓ and on f_ℓ separately and measure held-out AUC, repeating over 5 cross-validation seeds. A per-layer difference $\mathrm{AUC}(a_\ell) - \mathrm{AUC}(f_\ell)$ is counted as real only if its magnitude exceeds both 2σ (its own across-seed standard deviation) and a floor of 0.03. Table 4 shows the peak block.

Table 4: Truth AUC by component at the readability peak (block 15), CounterFact, mean $\pm$ std over 5 seeds. Residual = the summed stream (reference).

	residual	attention	FFN
AUC	0.792 ± 0.003	0.845 ± 0.003	0.734 ± 0.004

The alternation survives re-seeding and reproduces structurally across both datasets. On CounterFact, attention stably dominates readability at layers $\{6, 7, 10, 12, 13, 14, 15, 17, 18\}$ and the FFN at $\{8, 11, 16, 19, 20, 21, 22, 23, 25, 26\}$. At the peak (block 15), where the truth axis is read, attention leads by $+0.110 \pm 0.004$. **Interpretation (not yet causal):** attention appears to carry the truth signal most strongly through the middle band, including the peak. Readability, however, is not necessity, which Sec. 7.3 tests directly, and which changes the picture.

7.2 The expanded FFN space does not help

The FFN expands its input to 8960 dimensions, applies SwiGLU, and re-compresses it. One might expect the truth signal to be more cleanly readable inside that larger, putatively sparser space, before compression. We capture the expanded activation (the input to `down_proj`) and fit the truth axis there. In this $p \gg n$ regime the permutation null is elevated, so only the margin over the null counts: at the peak the expanded axis clears its null cleanly (curated 0.953 vs. null 95th percentile 0.669,

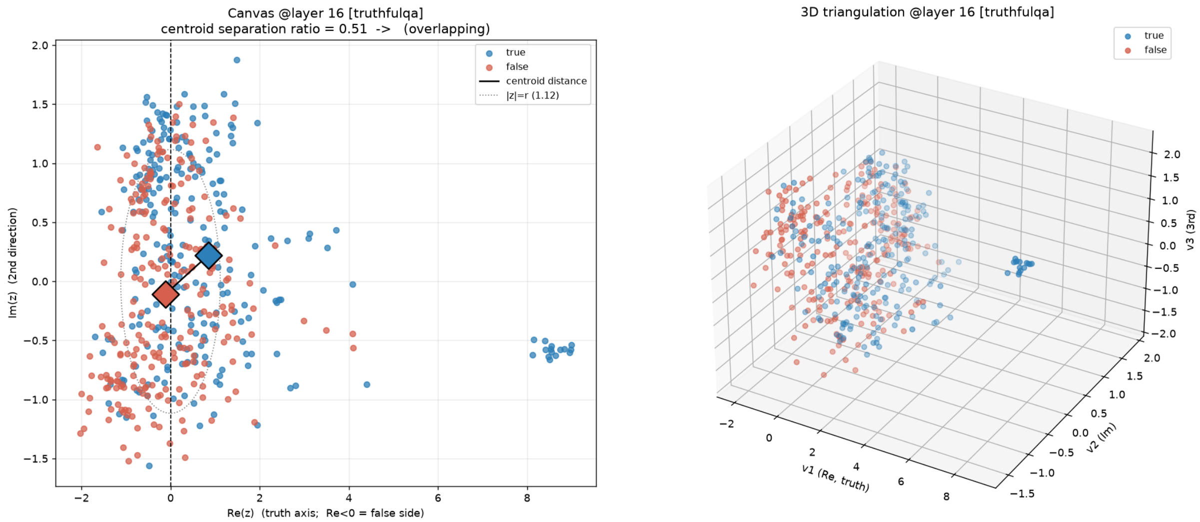


Figure 3: Experiment 7, TruthfulQA at layer 16. The same picture on more ambiguous facts: centroid-separation ratio 0.51 (overlapping), no three-dimensional clumps. Lower knowledge, lower separability, identical one-dimensional structure.

margin +0.284, $\hat{p} = 0.0099$; CounterFact 0.816 vs. 0.561, margin +0.254, $\hat{p} = 0.0099$). **But relative to the compressed residual at the same layer, the expanded space gains essentially nothing**: +0.017 (curated) and +0.023 (CounterFact). Raw architectural expansion does not undo superposition on these facts. The implication for future work is concrete: sparsity must be *imposed*, for example by a trained sparse autoencoder, not merely afforded by a wider space.

7.3 Causal ablation: attention builds, the FFN erodes

To separate readability from necessity we zero a component's contribution over a band of layers (a hook returns zeros, so $h + 0 = h$), let the effect propagate, and measure the collapse of truth AUC at a downstream readout (Table 5).

Table 5: Causal ablation on CounterFact (250 pairs). Drop = intact − ablated; positive drop = the component is causally necessary.

band (layers)	readout	intact	attention-off (drop)	FFN-off (drop)
11–15 (middle)	block 15	0.793	0.487 (+0.306)	0.819 (−0.026)
16–18 (transition)	block 18	0.712	0.708 (+0.004)	0.815 (−0.103)
19–26 (late)	block 27	0.537	0.534 (+0.003)	0.539 (−0.002)

Three readings, in order of confidence:

- **Middle band: attention is causally necessary.** Removing attention collapses truth from 0.793 to 0.487 (drop +0.306); removing the FFN does nothing (−0.026). Attention does not merely read the signal here, it builds it.
- **Transition band: the FFN *degrades* the signal.** Removing attention again does nothing (+0.004), but removing the FFN *raises* truth AUC from 0.712 to 0.815 (drop −0.103). From layer 16 onward the FFN injects content that lowers true/false separability. This is the first

causal support for an earlier conjecture, that the FFN adds context which amplifies the overlap on uncertain facts, though it rests on one band and one dataset and should be treated as a lead, not a result.

- **Late band: nothing remains to destroy.** The intact AUC is already about 0.54 (near chance): the truth signal has evaporated toward the output, so neither ablation has an effect.

This falsifies the initial "attention builds, FFN consolidates" hypothesis. There is no late consolidation; the FFN's apparent late *readability* dominance (Sec. 7.1) was the readability of a decaying signal, not active maintenance. The corrected picture: attention constructs the truth representation in the middle layers; the FFN begins to erode it in the transition band; by the late layers it is gone, as the model shifts from representing the fact to predicting the next token.

8 Discussion: what a single axis can and cannot read

It is tempting to view a cheap geometric probe as a low-cost version of dictionary learning, reading superposition without expanding it. We argue this is impossible, and that the impossibility is a capacity statement, not an implementation gap. A projection is an inner product: it collapses a d-dimensional state to a single scalar, discarding $d-1$ dimensions. From that scalar one cannot reconstruct which features were present, only how much a chosen direction was. "Reading the superposition" requires asking the question once per dictionary atom, which over an overcomplete dictionary of size m is exactly the $O(d\,m)$ sparse autoencoder. Compression (project to a scalar) and disambiguation (expand to a sparse code) are not inverse cheap-vs-expensive versions of one operation; they are different operations, and only the second reads superposition. The gaps of Sec. 5.2, 0.062 on homogeneous unknown material and 0.203 on the heterogeneous mix, are direct measurements of the truth information that a single axis structurally cannot return.

The consequence is a clear scope. This method's value is *structural and cheap*: a training-free axis that, on facts the model knows, rivals a trained probe at two dot products per token; a quantitative law relating the readability of truth to the model's knowledge of the fact; and a route to the polarity direction that needs no polarity labels. It is not a substitute for sparse autoencoders and should not be presented as one.

9 The axis reads truth and confidence

The seven experiments establish that there is no second coordinate of truth to find. This has a constructive consequence that is easy to miss, so we state it plainly. The same axis admits *two* readings, not one.

Its *sign* classifies a statement: $\mathrm{Re} > 0$ is the true side, $\mathrm{Re} < 0$ the false side. Its *separability*, how cleanly true and false split on a given set of facts, measures how well the model knows that material. The geometry that answers "is this statement true?" therefore also answers, indirectly, "does the model know this?", because a model that does not know produces overlapping rather than separated representations (Figs. 2, 3). These are not two axes; they are two readings of one axis: the direction of the arrow, and the sharpness of the arrow.

The reading is quantitative at the level of a distribution. The position-axis separation is 0.795 on CounterFact and 0.656 on TruthfulQA: by this measure the model "knows" the CounterFact material better than the TruthfulQA material, read off the geometry alone, with no external grading. We report this as a property of the dataset, not yet of the single fact. Calibrating a per-fact confidence, a local separability around one statement that predicts whether the model actually errs on it, is the natural next step, and requires a behavioral label (does the model answer the open question correctly?) that the minimal-pair design does not contain. We mark it as future work rather than a claim. Section 11 returns to this label and to a prerequisite it exposed.

10 Why unknown facts overlap: the signature of superposition

It remains to interpret *why* the classes overlap on unknown facts rather than simply asking *that* they do. The overlap is not measurement noise; it is the observable signature of unresolved superposition, and the explanation follows from the capacity argument of Sec. 8.

A projection onto a single direction returns a scalar: it collapses the d-dimensional state to one number and discards $d-1$ dimensions. When the model knows a fact, its truth content has been allocated a clean, dominant direction of its own (the residual stream has, in effect, dedicated an axis to it), and the single projection recovers almost all of it (position AUC 0.795, near the full-probe ceiling). When the model does not know a fact, there is no dedicated direction: the weak truth-relevant signal remains distributed across many superposed features, and no single axis can pull it out of the tangle. Recovering it would require asking the question once per dictionary atom, the expansion that a sparse autoencoder performs at cost $O(d\,m)$, which is a different operation from a single projection, not a cheaper version of it.

The consequence is exactly what the visual observations show. On known facts the truth occupies its own direction, the classes form two regions, and the centroid ratio is high. On unknown facts the truth is spread across the superposition, the classes overlap, and the centroid ratio falls below one. The dense, uniform overlap of true and false on unknown facts is therefore not a failure of the probe but a direct, visual readout of superposition that the probe structurally cannot resolve. The probe gaps of Sec. 5.2 and the low centroid ratio measured here are views of the same quantity: the truth information that lives off-axis on facts the model does not know (0.062), inflated further when heterogeneous material forces one axis to serve many truth frames at once (0.203). Subsequent work Part II begins to resolve that superposed remainder into named, signed components: what this paper can only call unresolved superposition acquires, there, a geometry.

11 A note for future work: verify behavior before geometry

Two natural extensions, a per-fact behavioral confidence label (Sec. 9) and a study of the geometry of *logical* (as opposed to semantic) truth, share a prerequisite that is easy to overlook, and we record it here as methodological guidance rather than as a result.

An informal behavioral check (not a controlled experiment). Before building any geometric study of a capability, one should confirm that the capability exists behaviorally. We ran a small, qualitative probe of the base model in open-ended completion (greedy decoding, no context carried between prompts). Two things were apparent on a handful of prompts. First, the model reproduces the common semantic facts used throughout this paper (capitals, chemical symbols, authorship, currencies), consistent with the "facts the model knows" assumption behind the curated pairs. Second, on symbolic transitive syllogisms built from nonce words ("a *grok* is heavier than a *blarn*; a *blarn* is heavier than a *fendu*; therefore ..."), the base 1.5B model does *not* perform the deduction: it deflects into multiple-choice or code scaffolding rather than completing the transitive conclusion. We stress that this is an informal observation on a few prompts, not a measured result, and we draw no quantitative claim from it.

The principle. Studying the geometry of a capability presupposes the capability. Measuring the hidden-state geometry of a task the model cannot perform yields noise, and that noise is easily mistaken for "the property is distributed/complex" when the truthful reading is "the property is absent." A cheap behavioral feasibility check should therefore precede any geometric study of *logical* truth; on this model that study is not feasible, which is itself useful to know in advance. The same principle applies to transfer: before studying the geometry of truth in a second model, one should first verify behaviorally that the second model knows the facts in question. (In the subsequent multi-model work of Part II this principle is applied systematically: the behavioral know-rate is measured before any geometry.)

A hypothesis to test, not a claim. The way the base model deflects on uncertain input, falling back on memorized *format* scaffolding (exam layouts, question/answer templates, code), suggests, but does not establish, that part of the overlap on unknown facts may carry a sentence-format component rather than pure superposition noise. This is consistent with the format confound that makes mixing datasets inadvisable (Sec. 12). We flag it as a direction for a future, properly controlled experiment, not as evidence.

12 Limitations

These results are confined to a single, small model (Qwen2.5-1.5B); the entire battery, the positive results and all seven falsification experiments, should be re-run at larger scale before the one-dimensionality is treated as general (the partial replications available so far are summarized, with their scope, in Part II). It is plausible that departments, dynamics, or a genuine second dimension emerge with scale, precisely because a 1.5B residual stream may be too compact to separate them. The datasets are small (tens to a few hundred pairs), so confidence intervals are wide despite the permutation control. Cross-validation is over pairs, not over topics: pairs from the same CounterFact relation may fall in different folds, so generalization to entirely unseen topics is not certified by the protocol of Sec. 4.4; the within- vs. cross-category transfer measurements of subsequent work Part II address the question directly. The "unknown-fact" interpretation of the CounterFact gap and of the low separability is plausible but confounded: a fact the model does not know and a fact whose truth is genuinely multidimensional are not separated here, and the low-separability reading is further confounded with sentence-format effects, which is why mixing datasets must be avoided. Disentangling ignorance from genuine multidimensionality (for example by filtering CounterFact to facts the model can answer, using the behavioral label of Sec. 11) is necessary to make the dimensionality claim quantitative. The anatomy results are Tier 2: margins in the expanded-space and transition-band analyses are modest and based on single datasets, the factual-knowledge label is noisy, and ablation measures necessity for *linear readability* of truth, not for the model's behavior. We use last-token pooling; other aggregations may shift the axis.

13 Summary of Part I

Stripped of a decorative complex-plane framing, the positive results survive on a real model under a falsifiable protocol: a training-free truth axis, identified without labels up to one global sign, carries a permutation-tested signal at cost $O(d)$; the gap between that axis and a full probe scales with how well the model knows the fact, suggesting a knowledge-dependent dimensionality of truth; and the difference SVD recovers the supervised polarity direction of [5] at cosine 0.959 without polarity labels. The one-dimensional reading withstands seven falsification experiments, spanning a second dimension, depth-global structure, dynamics, departments, energy, phase, and three-dimensional clustering, none of which exhibits a richer separating coordinate. Beneath the axis, an exact decomposition of the residual stream locates its origin: attention builds the signal in the middle layers, causally, while the FFN erodes it after the peak, and by the late layers the model has moved on from representing the fact to predicting the next token. We read a result that resists seven falsifications as strengthened, not weakened, interpret the residual overlap on unknown facts as the visual signature of unresolved superposition, and regard the axis as a light, honest, almost-free structural probe to sit alongside generation. Natural next steps, partly realized in the subsequent work summarized in Part II, are the multi-model study of the dimensionality law, a knowledge-controlled dataset built from a behavioral label, and the category-level structure of the axis, each preceded by a behavioral feasibility check of the kind described in Sec. 11. All experiments in this manuscript are fully reproducible; the exact execution commands for both parts are consolidated at the end of the text.

Part II: Mechanics and Geometry Across Two Families

14 Scope and evidence levels

Results are organized in three evidential tiers, continuing the Tier 1 / Tier 2 separation of Part I with a finer grain, and never mixed. **Tier A**: replicated across models with predictions written before the runs. **Tier B**: causal but on fewer models, or with multiple independent methods on one. **Tier C**: leads, declared as such (Sec. 24). Retractions and failed predictions are reported in the text at the point where they occur; the largest, the correction of our own flip interpretation, has its own section (17.2).

15 Setup

Four base (non-instruct) models across two families:

Table 6: Models. The truth peak p is the block whose intact residual maximizes held-out truth AUC (protocol of (Part I)); "block b" denotes the output of transformer block b. Know-rate: fraction of 200 CounterFact prompts whose true target the model reproduces under greedy decoding (seed 0), measured before any geometry. Axis and probe AUC: CounterFact, 250 pairs, held-out; gap = probe − axis.

model	blocks	d_{model}	peak p	know-rate	axis	probe	gap
Qwen2.5-1.5B	28	1536	15	30.0% (61/200)	0.792	0.854	0.062
Qwen2.5-3B	36	2048	16	37.0% (74/200)	0.832	0.881	0.049
Llama-3.2-1B	16	2048	7	35.5% (71/200)	0.690	0.831	0.141
Llama-3.2-3B	28	3072	9	41.0% (82/200)	0.785	0.897	0.112

Two regularities and one boundary, extending the dimensionality claim of (Part I): within each family the gap decreases as the know-rate increases; across families a family-level offset remains, with Qwen concentrating more of the truth signal on the single axis at comparable knowledge. An earlier version of this table imported the Qwen2.5-1.5B row from the first part's mixed-dataset column while the other rows were measured on CounterFact alone, a protocol inhomogeneity caught by re-measurement; all rows above come from one command under one protocol, and the earlier phrasing that the gap "tracks knowledge every time" is retired in favor of the two-part statement.

Dataset: `NeelNanda/counterfact-tracing` (pinned revision `c945b08...`, loaded as Parquet only), 250 minimal pairs unless stated. Decomposition runs use `float32` and are gated by the additive identity check of (Part I) (median relative errors 0.65–1.23 $\times 10^{-7}$ across the runs of this paper); extraction-only runs use `bfloat16`. The spread of the four peaks in relative depth (0.32–0.54) already anticipates the negative of Sec. 20: no fixed depth, absolute or relative, locates truth.

16 Methods

16.1 Frames and the fixed-frame contribution statistic

Within a block the residual stream updates additively, $h_{b+1} = h_b + a_b + f_b$, with a_b the attention contribution and f_b the FFN contribution, captured by forward hooks (Part I).

Definition 1 (Truth frame)**.** *The* post frame F_b^{post} *is the oriented truth axis v_1 fit (on training folds only, true side positive) on the block output $h_{b+1} = h_b + a_b + f_b$; it contains f_b and a_b. The* pre frame F_b^{pre} *is the same construction on the pre-FFN state $h_b + a_b$; it contains a_b but excludes f_b. A frame* exists *at b when the axis fit there clears its permutation null held-out.*

The statistic for a contribution c at layer L measured against a frame F with axis v_1 is the held-out class gap and its effect size,

$$\mathrm{gap}(c;F) \;=\; \overline{(v_1\cdot c)}\big|_{\text{true}} - \overline{(v_1\cdot c)}\big|_{\text{false}}, \qquad d'(c;F) \;=\; \frac{\mathrm{gap}(c;F)}{\text{pooled std}}, \tag{4}$$

with the pooled standard deviation over the held-out projections of both classes. A negative FFN gap means the FFN literally writes toward the false side of the frame. Stability criterion, fixed in code before the consolidation runs: a value is *stable* if its magnitude exceeds both 2σ across 5 cross-validation seeds and a floor of 0.03.

16.2 The exact SwiGLU split

The FFN computes $f = W_{\text{down}}(g \odot u)$ with $g = \mathrm{silu}(W_{\text{gate}}x)$, $u = W_{\text{up}}x$. The projection of f on v_1 is $w \cdot (g \odot u)$ with $w = W_{\text{down}}^{\top} v_1$, the residual axis pulled back through the down-projection; no axis is ever fit in the expanded space, avoiding the $p \gg n$ regime (more dimensions than samples) entirely. With the symmetric intra-pair split ($\Delta g = g_t - g_f$, $\bar{g} = (g_t + g_f)/2$, likewise for u) the mixed term cancels and the identity is exact:

$$w \cdot (g_t \odot u_t - g_f \odot u_f) = \underbrace{w \cdot (\Delta g \odot \bar{u})}_{\text{gate term}} + \underbrace{w \cdot (\bar{g} \odot \Delta u)}_{\text{value term}}. \tag{5}$$

Verified numerically at 7.4×10^{-8} relative error on real data.

16.3 Interventions

Ablating a component means replacing its additive contribution (f_ℓ or a_ℓ) with the zero vector through a forward hook, for every layer ℓ in a band (peak+1 to peak+3; readout at the band end, on every model), during the forward pass: the weights are untouched, the module still computes and its output is discarded, and the residual update reduces to $h_{\ell+1} = h_\ell + a_\ell$ (for FFN ablation) or $h_{\ell+1} = h_\ell + f_\ell$ (for attention ablation). Two refinements. (i) A *frozen control* obtained for free: with both components zeroed over the band, the readout state is identically the band-entry state, so the frozen reference is read off the intact pass. (ii) *Gate-freeze and value-freeze*: at the reading position, g (respectively u) is replaced by its intra-pair mean cached from the intact run, killing the class-driven variation of one stream while the FFN stays on.

On the validity of the freezes. Any activation-level intervention risks pushing the network off its data manifold, so three properties of this design matter. The replacement value is the *intra-pair mean*: a convex combination of the two real activations of two nearly identical inputs, which is as close to on-manifold as an intervention of this kind gets. The two freezes are symmetric by construction, so their *contrast* (value recovery +0.064 to +0.123 against gate recovery +0.008 to +0.056 across the four models) controls for generic disruption: an off-manifold artifact has no reason to respect the gate/value distinction. And the interventional verdict agrees with the purely observational exact split of Eq. 5, which touches nothing. The residual risk is declared, not dismissed.

16.4 Static circuit norms

Per attention head (GQA-aware), the gauge-invariant circuit norms $\|W_q^\top W_k\|_F$ (routing) and $\|W_o W_v\|_F$ (writing) [9]. Individual weight norms are gauge-dependent ($W_q \to \alpha W_q$, $W_k \to W_k/\alpha$ leaves the function unchanged); the products are not. A gauge self-test gates the computation. The *crossover* is the first layer from which the median-normalized OV/QK ratio stays above 1 for at least three consecutive layers, a criterion fixed in code before any run.

16.5 The axis-provenance control

The objection this method answers, raised by the author against the author's own instrument, is that a result anchored to "the block where the ruler was fit, plus one" could be a property of the ruler rather than of the FFN. The control: fit frames at several blocks $b \in \{p-4, p-2, p, p+2\}$, both post and pre, and recompute the full d' profile of each component under each frame. The discrimination criterion was fixed before the runs and validated on synthetic ground truth: a planted global axis with a planted writer flip stays pinned under every frame; a planted circular writer collapses under the pre frame (collapse ratio 0.04) while planted genuine alignment survives (0.99). The released tools print full matrices with no automatic verdict; the reason is a repeated methodological lesson recorded in Sec. 20.

16.6 Per-category axes, transfer, decoding, and the Mantel statistic

CounterFact is grouped by Wikidata relation; the top-K relations by unique pairs are used (K=8 for the registered falsification, 60 pairs each): P103 mother tongue, P1412 languages spoken, P176 manufacturer, P27 citizenship, P30 continent, P37 official language, P413 sport position, P495 country of origin. Per-category truth axes are fit at the peak block, each oriented within its own category, so the sign of a cross-category cosine is information: negative means the shared direction reads the other category's truth backwards. Four measurements: (i) signed cosine matrices at the peak and at an early block (block 2, the lexical surface control); (ii) the held-out *transfer matrix*, cell (i, j) being the held-out AUC of category i's axis on category j's pairs; (iii) nearest-centroid *decoding* of the category from the FFN's class-signed write direction Δf at peak+1, against the same decoder on early residual differences (the lexical baseline); (iv) the *Mantel test*, a permutation test for the correlation between two distance matrices, here applied under category-label permutation to compare the two families' signed-cosine arrangements.

17 First movement: the flip, its correction, and the relational law

17.1 The peak-relative flip (the measurement that survives)

For each model the prediction was written before the run: under the peak-fit post frame, the FFN's gap is pro-truth at the peak block p and flips to stable anti-truth at $p+1$.

Table 7: The flip at peak+1 under the peak-fit post frame. CounterFact 250 pairs, 5 seeds, permutation null with 100 within-pair swaps ($\hat{p} = 0.0099$ on every row). All four rows from the uniform July 10 consolidation; the Qwen2.5-1.5B row supersedes the earlier single-seed scan (+0.728/−0.692), with which it agrees to the second decimal.

model	$\text{gap}_{\text{ffn}}(p)$	$\text{gap}_{\text{ffn}}(p+1)$	$d'(p)$	$d'(p+1)$
Qwen2.5-1.5B	$+0.730 \pm 0.002$	-0.691 ± 0.002	+0.58	−0.82
Qwen2.5-3B	$+0.575 \pm 0.005$	-0.926 ± 0.002	+0.68	−1.25
Llama-3.2-1B	$+0.092 \pm 0.001$	-0.111 ± 0.001	+0.54	−0.94
Llama-3.2-3B	$+0.087 \pm 0.001$	-0.167 ± 0.001	+0.49	−1.26

The four peaks sit at different absolute and relative depths (Table 6); the flip sits at peak+1 in all four. Aligned in peak-relative coordinates and normalized by the flip amplitude, the per-layer profiles collapse onto a common shape: rise into the peak, spike at +1, decay toward zero (Fig. 4). The measurement is correct and reproduces; its interpretation, that the flip is anchored to the peak, is what the next subsection tests.

17.2 The correction: the control bites its owner

Under frames fit at $p-4$, $p-2$, p, $p+2$ (Sec. 16.5), the transition does not stay at $p+1$: it tracks the frame. On every model the post-frame diagonal reads pro at b and anti at $b+1$ for every b where a frame exists (Table 8). The peak-anchored flip was the $b=p$ slice of a relational pattern present at every such block. An accidental replication is worth recording: a consolidation run launched with a wrong default re-measured Qwen2.5-3B with the frame at block 15, off its peak, and returned the same pro/anti pair at 15/16 (+0.541/−0.545, 5 seeds, $\hat{p} = 0.0099$); the accident confirms the law at a frame no one chose.

Table 8: Post-frame diagonals: per fit block b, the FFN's held-out d' at b and at $b+1$. Pro at b, anti at $b+1$, wherever a frame exists.

model (peak)	frame b	$d'_{\text{ffn}}(b)$	$d'_{\text{ffn}}(b+1)$
Qwen2.5-1.5B (15)	13 / 15 / 17	+0.61 / +0.57 / +0.52	−0.79 / −0.82 / −0.73
Qwen2.5-3B (16)	14 / 16 / 18	+0.06 / +0.69 / +0.73	−0.09 / −1.23 / −0.79
Llama-3.2-1B (7)	5 / 7 / 9	+0.13 / +0.54 / +0.19	−0.07 / −0.95 / −0.54
Llama-3.2-3B (9)	7 / 9 / 11	+0.37 / +0.49 / +0.57	−0.42 / −1.25 / −1.39

Under the pre frame the FFN's "pro at own block" term does not merely shrink; it inverts sign (Table 9). Measured against the frame that exists at the moment it writes, the FFN is strongly anti-aligned. The pro term of the post frame was, in substantial part, the frame measuring its own ingredient: f_b is inside h_{b+1} and drags the axis fit there toward itself. Where the axis is weak or absent, the pattern correctly vanishes: frames fit before the Qwen-3B ignition (fit-block cosines to the peak axis 0.049 and 0.079) and before the Llama-1B peak return noise. What the failure bought: a general law instead of a special case, and a live demonstration that the safeguards bind their owner.

17.3 The relational law

Table 9: Pre-frame diagonals: the same cells with f_b excluded from the frame's data. The FFN inverts sign; attention, measured identically, is the mirror image.

model	frame b	$d'_{\text{ffn}}(b)$	$d'_{\text{attn}}(b)$
Qwen2.5-1.5B	13 / 15 / 17	−0.63 / −0.41 / −0.78	+1.56 / +1.31 / +1.04
Qwen2.5-3B	14 / 16 / 18	−0.14 / −1.16 / −0.46	+0.26 / +1.38 / +1.40
Llama-3.2-1B	7 / 9	−1.17 / −0.82	+1.26 / +0.82
Llama-3.2-3B	7 / 9 / 11	−0.81 / −1.43 / −1.41	+0.99 / +1.63 / +1.75

The decisive zone is the sub-diagonal of the pre matrices: contributions at layer L measured against frames of blocks strictly before L, which contain neither f_L nor a_L and are clean of every containment effect. There (Fig. 5) the FFN is negative and attention positive nearly everywhere a frame exists, on all four models. At the peak, under the same clean frame, the two components are near mirror images: $+1.31/-0.41$ (Qwen-1.5B), $+1.38/-1.16$ (Qwen-3B), $+1.26/-1.17$ (Llama-1B), $+1.63/-1.43$ (Llama-3B). Pre-registered predictions for the two Llamas, written before those runs, were all confirmed. The regularity can be stated compactly.

Empirical Law 1 (Propagate vs. overwrite)**.** *Let F be a truth frame that exists at block b, and let $L \geq b$ be a layer whose contributions are not contained in F (any L for pre frames at $b = L$; any $L > b$ for both frame types). Then, as an empirical regularity across the four models studied: $d'(f_L; F) < 0$ and $d'(a_L; F) > 0$ wherever both are stable. Attention propagates frames it did not write; the FFN opposes them and deposits the material of later frames.*

The law dissolves the earlier implicit picture of an axis being eroded like an object: the axis is not an object in the model but a moving statistical summary of where the class difference lives, and the FFN's class-signed writes are what successive frames are made of. This is the dynamical face of the key-value write-head physiology attributed to MLPs [12], and a candidate microscopic writer for the macroscopic finding of [15].

18 The erosion: real, frame-free, and carried by the value stream on four models

The correction separates two phenomena the peak-anchored reading conflated: frame-relative rotation (what the flip was) and frame-free decay of class signal after the peak (the genuine erosion). The erosion stands on measurements that use no frozen frame, now on all four models (Table 10); the predictions for the Llama rows were registered before those runs and confirmed. This section is Tier A.

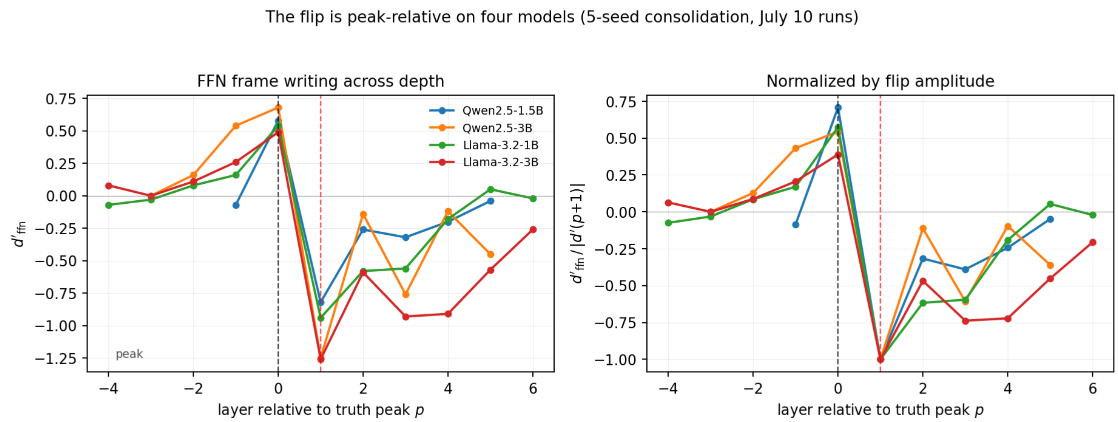


Figure 4: The FFN's frame writing (d') across depth, aligned at the truth peak, four models. Right: normalized by the flip amplitude.

Table 10: Causal ablation over the band peak+1 to peak+3, readout at band end; held-out refit truth AUC. FFN-off restores signal on every model; attention-off is neutral on the smaller models and destructive on both 3B models (the tug-of-war of Sec. 19).

model	intact	FFN-off	attn-off	frozen
Qwen2.5-1.5B	0.712	0.815	0.708	0.793
Qwen2.5-3B	0.793	0.846	0.738	0.831
Llama-3.2-1B	0.551	0.744	0.531	0.691
Llama-3.2-3B	0.740	0.847	0.593	0.785

Table 11: Inside the SwiGLU, four models. Left: the exact split of Eq. 5 at the flip layer (peak-fit frame). Right: causal freezes over the band (AUC recovery vs. intact). The anti-truth write and its causal carrier are the value stream everywhere; the gate is neutral or mildly pro-truth.

	split at flip			freeze recovery	
model	gate	value	total	gate	value
Qwen2.5-1.5B	+0.041	−0.734	−0.692	+0.040	+0.089
Qwen2.5-3B	−0.120	−0.806	−0.926	+0.008	+0.064
Llama-3.2-1B	+0.016	−0.127	−0.110	+0.028	+0.123
Llama-3.2-3B	+0.021	−0.186	−0.166	+0.056	+0.122

Three annotations keep the table honest. First, the earlier hypothesis that the gate carries the erosion (which neurons open switching the stream toward next-token context) is falsified in the same direction on every model; at the peak, by contrast, the split shows the gate carrying about two thirds of the pro-truth construction (Qwen-1.5B: gate +0.486, value +0.242). Second, on Llama-3.2-3B the value-freeze lands at 0.862, above the frozen reference 0.785: with the value stream's class variation silenced, the band's attention adds signal, which is the tug-of-war of the next section seen causally. Third, the earlier lead "FFN-off beats frozen", demoted to noise on the Qwen models (+0.022/+0.015, both under the stability floor), resurfaces above the floor on both Llamas (+0.052/+0.062); the demotion was correct for Qwen, and the effect is recorded as family-dependent, without a dedicated

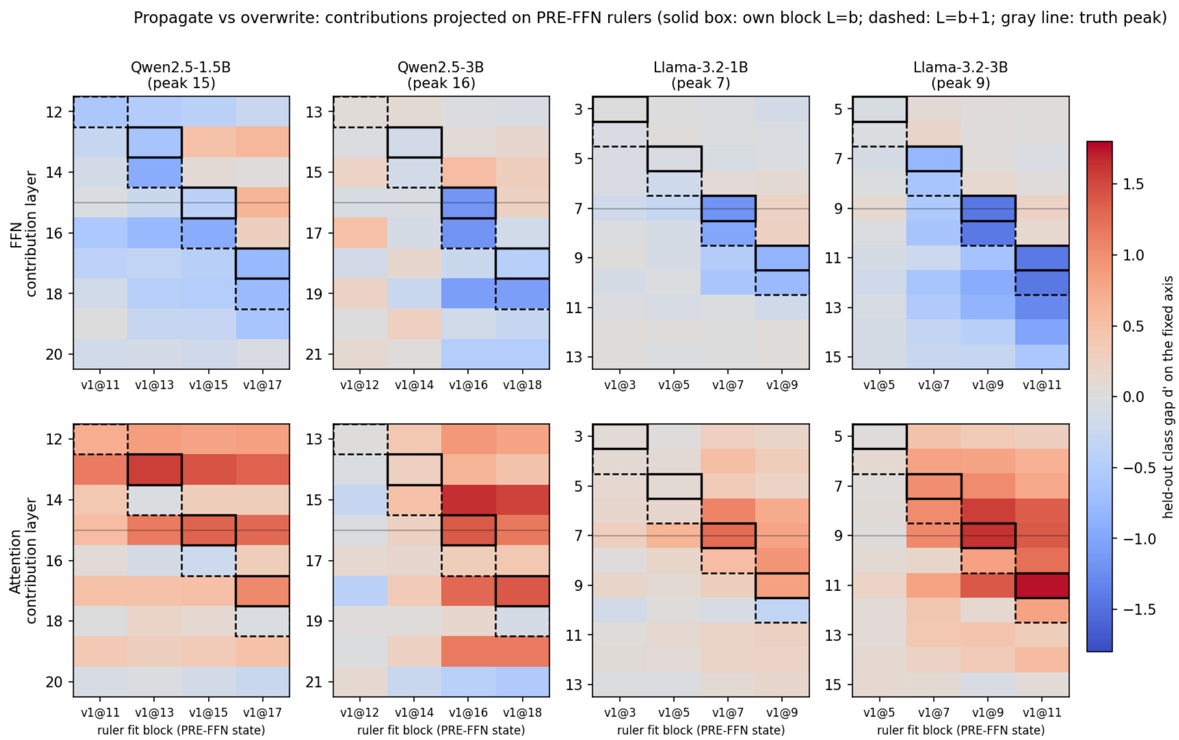


Figure 5: Propagate vs. overwrite on four models: held-out d' of each component's contribution (FFN above, attention below) against pre frames fit at four blocks. The FFN contribution is negative wherever a frame exists; the attention contribution is positive.

null, hence an observation and not a claim. A frame-free rotation check closes the projection-artifact objection on three models: the cosine of the mean FFN pair-difference with v_1 jumps sign in a single step at peak+1 with the write magnitude intact (deltas -0.741, -0.654, -0.632 on Qwen-1.5B, Qwen-3B, Llama-1B), where genuine rotation predicts a smooth multi-layer sweep.

19 Two traits, discriminated: tug-of-war is scale, ignition is family

Two pre-registered discriminations with opposite outcomes, which together show that scale and training recipe leave distinct, measurable signatures.

Tug-of-war is a trait of scale. In the erosion band, removing attention is neutral on the two smaller models (refit change -0.004 on Qwen-1.5B, -0.020 on Llama-1B, both under the floor) and destructive on both 3B models (-0.055 on Qwen-3B, -0.147 on Llama-3B). The registered discrimination read: present on Llama-3B and absent on Llama-1B implies scale; absent on both implies a Qwen-family trait. Outcome: scale. In the larger models the post-peak band is a tug-of-war in which attention still writes pro-truth while the FFN's value stream overwrites, not a passive decline.

Ignition is a trait of the Qwen family. The Qwen-3B per-layer AUC curve switches on in a single layer ($0.526 \to 0.790$), driven by a coordinated two-layer construction burst; the Qwen-1.5B curve is a smooth ramp. The registered discrimination: scale predicts a step in Llama-3.2-3B; family predicts a ramp. Measured: a ramp ($0.526 \to 0.657 \to 0.664 \to 0.749 \to 0.785$ across levels 6–10). Ignition is, on present evidence, a trait of the Qwen recipe at the 3B scale, not of scale itself.

20 A pre-registered negative, cross-family: the static crossover does not track the peak

Motivated by an external observation that value-to-query weight norms scale with depth [16], the gauge-invariant version was tested: does the layer where the writing circuit overtakes the routing circuit predict the truth peak? The prediction for the Llama models, registered before their runs, was that it does not. Outcome (Table 12): the crossover sits at a stable late relative depth on all four models (0.61–0.75) while the peak wanders (0.32–0.54); the two coincide only on Qwen2.5-1.5B, the accident that spawned the hypothesis the first working note falsified across scale. The depth-scaling of the circuits may be real and general; it is not where truth lives. The late-layer OV explosion (Qwen-1.5B sums 338.9/322.0 at layers 26–27) instead matches the decoupling of Sec. 24.

Table 12: The OV/QK crossover (first layer sustaining ratio > 1 for three consecutive layers) against the truth landmarks, four models. Distance is to the nearer of peak and flip.

model	crossover	peak / flip	distance	rel. depth (cross vs. peak)
Qwen2.5-1.5B	17	15 / 16	1	0.61 vs. 0.54
Qwen2.5-3B	24	16 / 17	7	0.67 vs. 0.44
Llama-3.2-1B	12	7 / 8	4	0.75 vs. 0.44
Llama-3.2-3B	17	9 / 10	7	0.61 vs. 0.32

A methodological note on automatic verdicts. During the regeneration of this table, the circuit tool's hardcoded landmark annotations (written for one model) mislabeled the truth peak on the other three and printed a spurious coincidence verdict for Llama-3.2-3B. It was the third automatic verdict to misread its own numbers in this project's history, after two in the provenance tooling. All released tools now print matrices and distances with no verdict prose; the reading belongs to the researcher. The raw norm tables, which depend only on the weights, were unaffected.

21 Second movement: the axis is a semantically signed mixture

If the FFN writes truth along directions that depend on what kind of fact is being written, the single axis is a mixture, and its ingredients can be measured. Design and statistics as in Sec. 16.6; the registered falsification runs on the two 3B models. Llama-3.2-1B is excluded from this section by a scoping rule stated in advance: it is the weakest-axis model of the four (axis AUC 0.690), and per-category axes fit on 60 pairs there risk reporting noise as geometry.

The signed geometry. At the peak, the language-relation block coheres (Qwen-3B cosines +0.27 to +0.76; Llama-3B +0.55 to +0.69), continent anti-aligns with that block (Qwen −0.07 to −0.20; Llama −0.09 to −0.15), sport-position is near-orthogonal to everything, and country-of-origin sides with the block (strongly on Llama, +0.51 to +0.75). The anti-alignment is confirmed by an independent statistic: transfer from the continent axis into the language block falls below chance (Llama: AUC 0.198–0.33); on Qwen the below-chance transfer is bidirectional, on Llama one-directional, a partial replication reported as such. Within-category transfer exceeds cross-category transfer on both models (Qwen 0.662 vs. 0.596; Llama 0.898 vs. 0.808), which also bears on the topic-generalization question raised for the protocol of Part I.

The reorganization with depth. The early-block matrices are not flat: they carry shared corpus statistics (the two families' early matrices correlate at $r = +0.934$). Depth does not create arrangement from nothing; it re-files it. Citizenship sides with geography at the embedding level and with the language block at the peak, on both families. The extreme case: the axes of languages-spoken and official-language start near anti-parallel (cos $= -0.90$ Qwen, -0.86 Llama, presumably a lexical opposition in how the two templates use language tokens) and merge at the peak (+0.76, +0.62). Within each model, the early and peak arrangements are uncorrelated or anti-correlated ($r = -0.547$ Qwen, -0.175 Llama): the peak structure contradicts, rather than inherits, the lexical surface.

The FFN writes truth along category-dependent directions. Decoding the relation from the unit write direction Δf at peak+1: 74.17% (Qwen-3B) and 81.67% (Llama-3B) against chance 12.5% and permutation null 95th percentiles near 15.5% ($\hat{p} = 0.0099$ both); lexical baseline 39.17% and 42.08%. Read together with Law 1: the component that overwrites truth frames does so along category-dependent directions. The mixture is not only in the representation but in the writing.

The registered falsification and its verdict. Five predictions were written before any Llama measurement of this geometry; outcomes in full, partials included. **R1** (language block on Llama, cosines in $[+0.15, +0.45]$): structure confirmed, magnitude outside the registered window on the high side. **R2** (continent anti-aligned, below-chance transfer both directions): signs confirmed; bidirectionality held only on Qwen. **R3** (Δf decoding at least 10 points over lexical): confirmed with a margin near 40 points. **R4** (the citizenship early-to-peak switch replicates): confirmed, plus the stronger merge. **R5** (the arrangement law): confirmed; off-diagonal peak matrices, Qwen-3B vs. Llama-3B, Pearson $r = +0.770$ (Spearman +0.893), Mantel null 95th percentile +0.498, $p = 0.0009$; robust to excluding the anomalous relation P176 ($r = +0.808$, $p = 0.0037$).

The arrangement law. Combined with the within-model reorganization, the reading is: the two families share a lexical starting arrangement, abandon it, and converge on the same functional arrangement. Same start, same move, same destination, different machines. The arrangement of the truth mixture is a property of the knowledge domain, not of the training recipe: a signed, truth-specific instance of cross-model representation convergence [17]. In hindsight this section also explains a Part I measurement from the other side: forcing one axis onto heterogeneous material costs 0.203 of held-out AUC against a full probe, versus 0.062 on homogeneous material (Part I), because the single axis is being asked to summarize a signed mixture with one number.

Anomaly, declared. On Llama-3B, relation P176 (manufacturer) shows within-CV and incoming transfer of 1.000 from every axis: universal perfect separability, which suggests surface structure in that relation's pairs rather than truth geometry. It is excluded in the R5 robustness check (which holds) and flagged for inspection before any claim rests on it.

22 Stress tests of the arrangement law: sign gauge, sample density, and the knowledge gate

The arrangement law of Sec. 21 was established at one scale: $K{=}8$ relations, 60 pairs each, one sampling seed. This section subjects it to three independent stresses, category count ($K{=}33$), sample

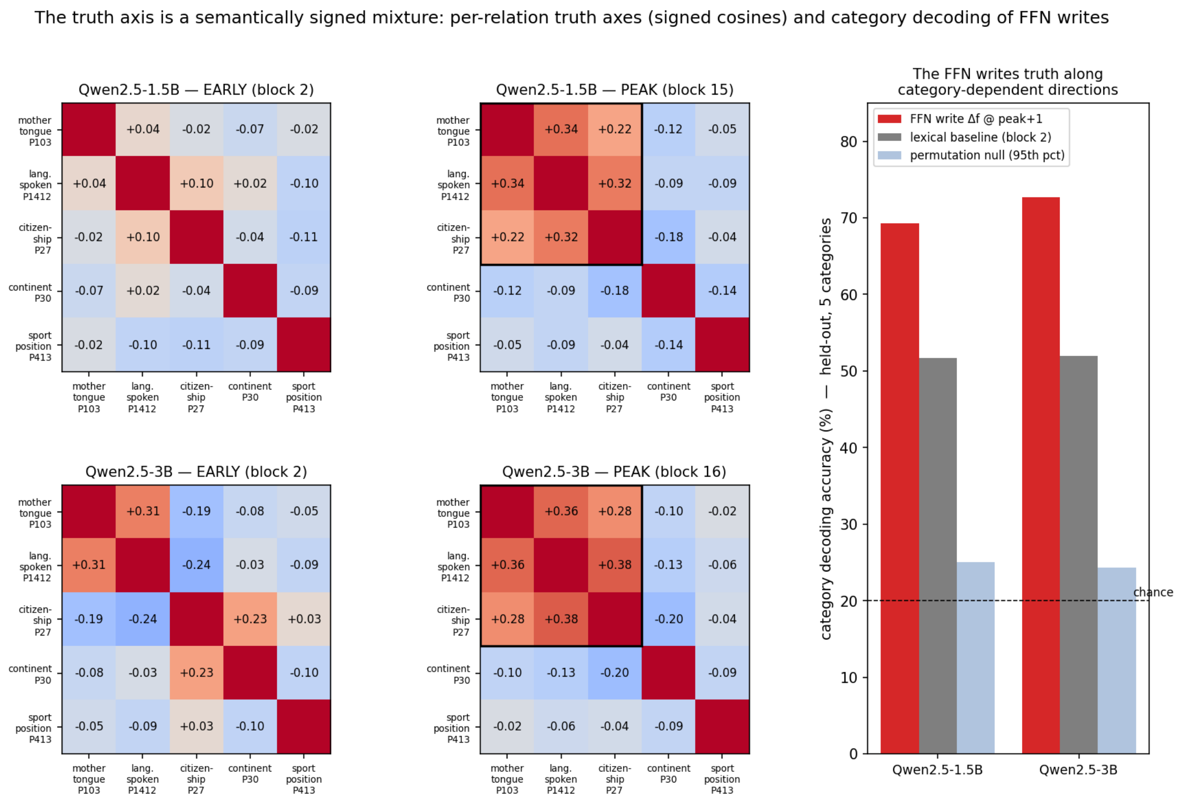


Figure 6: The category geometry: early vs. peak signed-cosine matrices, and decoding vs. the lexical baseline.

density (888 pairs per relation), and the seed of the pair sampling (six additional seeds), and reports, as results, one instrument failure discovered on the way and the correction that repairs it. All bundles are produced by one exporter under the pinned dataset revision, and every comparison is one command on two exported dictionaries.

The statistics, formally. Let $\hat{v}_1, \ldots, \hat{v}_K$ be the unit per-category axes of one model at its peak, $C_{ij} = \hat{v}_i \cdot \hat{v}_j$ their cosine matrix, and $s \in \{\pm 1\}^K$ a sign assignment (a *gauge*); the signed matrix under s is

$$M \;=\; D_s\, C\, D_s, \qquad D_s = \operatorname{diag}(s), \qquad M_{ij} = s_i s_j\, C_{ij}. \tag{6}$$

Three gauges appear below. The *per-category* gauge takes s_i from the true-side positivity of category i on its own pairs. The *global-axis* gauge takes $s_i = \operatorname{sign}(\hat{v}_i \cdot t)$ with t the truth axis fit on the pooled pairs. The *consensus* gauge, adopted in this section, takes

$$s^\star = \operatorname{sign}(u), \qquad u = \arg\max_{\|x\|=1} x^\top C\, x, \tag{7}$$

the leading eigenvector of C: the spectral relaxation of $\max_s \sum_{ij} s_i s_j C_{ij}$, i.e. the assignment that maximizes collective sign agreement. The per-category margin is $|u_i|/\|u\|_\infty$; components below 0.10 are declared unsigned rather than forced, and the overall sign of u is fixed by majority. Agreement between two models is the Mantel statistic

$$r = \operatorname{Pearson}\big(\mathcal{O}(M^A), \mathcal{O}(M^B)\big), \qquad \hat{p} = \frac{1 + \#\{b : r_b \geq r\}}{B+1}, \tag{8}$$

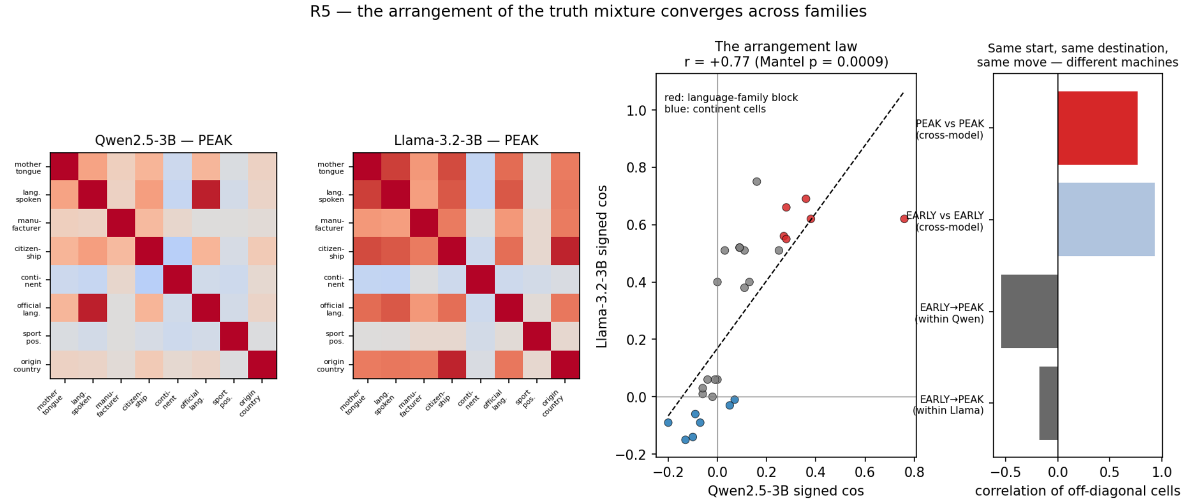


Figure 7: The arrangement law (R5). Left: the two K=8 peak signed-cosine matrices. Center: their off-diagonal cells against each other. Right: cross-model agreement at the peak and at the embedding level, and the within-model reorganization between them.

with $\mathcal{O}(\cdot)$ the off-diagonal cells and r_b obtained by jointly permuting the rows and columns of one matrix under a random category relabeling ($B = 9999$). The knowledge-restricted variant recomputes r on $\mathcal{C}_\tau = \{\, i : W_i^A \geq \tau \wedge W_i^B \geq \tau \,\}$ with W_i the within-category held-out AUC and $\tau = 0.6$. Per-category agreement is the *adherence* $\mathrm{adh}_i = \mathrm{Pearson}\big(\mathcal{O}_i(M^A), \mathcal{O}_i(M^B)\big)$ over the off-diagonal cells of row i, and the knowledge gate is the rank correlation of adh with a per-category knowledge measure. Finally, the *sign repair* used in diagnosis searches s by greedy coordinate descent to maximize $\mathrm{Pearson}\big(\mathcal{O}(D_s M^A D_s), \mathcal{O}(M^B)\big)$; the flips it reports name the unstable axes. It is a diagnostic and never a gauge: fitting signs to the reference under comparison would assume the conclusion.

The instability: per-category orientation is a thin gauge. Each axis is identified by Lemma 1 up to sign, and the sign was fixed within the category. That orientation margin is thin exactly where the model knows little, and a thin margin is a coin. The seed study exposed the coin: re-sampling the 60 Qwen pairs under six seeds left the unsigned structure $|C|$ essentially unchanged but produced sign storms in half the runs (three of six signed matrices anti-correlated with the canonical one, Mantel $-0.42/-0.26/-0.21$; sign repair restores $+0.46/+0.59/+0.73$, identifying coordinated flips of two to three axes). The same pathology governs density: at 888 pairs the full cross-family comparison collapses to $+0.233$ ($p = 0.15$), every category subset containing P495 fails while the same test excluding P495 returns $+0.621$ ($p = 0.005$), and the repair diagnostic identifies P495 as the flipped axis. The low know-rates of the affected relations place the instability where the dimensionality relation of Part I predicts weak, diffuse signal.

A failed cure, reported: the pooled global axis is capturable. The first repair we registered was the global-axis gauge. The control that killed it is worth the space it takes: on the Qwen K=8 pool the stored global direction has cosine 0.99 with the axis of P30 (continent), the one category that anti-aligns with the language block. The dominant direction of a heterogeneous pool is a property of the mixture: whichever category contributes the largest, most coherent difference vectors captures it, so the referee inherits exactly the problem it was appointed to judge. On Llama, whose pool is dominated by a compact high-knowledge block, the same estimator is healthy; the failure is knowledge-dependent, like everything else in this series.

The cure: the spectral consensus. The consensus gauge of Eq. 7 uses no pool. Under it the margins are legible on both families: the Llama block sits at +0.73 to +1.00 with P30 negative on every bundle (its semantic sign, not an error) and P413 near zero (the loner, reported unsigned); on the weakest bundle in the study (Qwen, 60 pairs) the block is positive with P30 at −0.28. Where the two gauges can be compared on a healthy pool (K=33) they flip identical category sets on both models, so the correction changes nothing where nothing was broken. A gauge, finally, cannot manufacture structure: only relative signs are observables, since any product of cosines around a closed cycle ($C_{ij}C_{jk}C_{ki}$) is invariant under every sign assignment. The anti-alignment of P30 survives the consensus precisely because its cycles with the language block are frustrated and no assignment can synchronize them; the gauge does not create that obstruction, it exposes it. And because the consensus is computed per model, independently, cross-family agreement under it cannot be an artifact of the gauge: two referees who never speak cannot collude.

Table 13: The arrangement law under the consensus gauge. Seed rows: Qwen2.5-3B, canonical seed vs. each of six re-samplings, K=8, 60 pairs. Scale rows: Llama-3.2-3B vs. Qwen2.5-3B. "Per-category" is the original within-category orientation, where measured. Mantel r on off-diagonal signed cosines; p from category-label permutation.

comparison	per-category gauge	consensus gauge	p (consensus)
seed 1 vs. canonical		+0.589	0.0011
seed 2		+0.643	0.0048
seed 3	3 of 6 negative	+0.354	0.0433
seed 7	(−0.42 to +0.86)	+0.811	0.0002
seed 8		+0.452	0.0328
seed 9		+0.681	0.0057
cross-family, K=8, 888 pairs	+0.233 (p=0.15)	+0.424	0.0034
cross-family, K=33, 60 pairs	+0.559	+0.559	0.0001
restricted to 19 known relations		+0.744	0.0001

The law at scale, and the knowledge gate. With the gauge repaired, the three stresses return one coherent picture (Table 13). The seed storms disappear: six of six comparisons positive and significant. The control arm was registered as a prediction of the gating itself: if the instability is knowledge-dependent, the high-knowledge family must be stable under any gauge. Six re-samplings on Llama-3.2-3B return Mantel +0.912 to +0.991 under the consensus gauge and +0.930 to +0.992 under the original per-category orientation (all $p \leq 0.0032$), with the same flip set on every bundle (P30 negative, the loner P413 unsigned at margins $|u_i| \leq 0.08$ across all seeds): the disease never touched the family whose margins are fat, which is what a knowledge-gated instability must do. On the same family the adherence-knowledge correlation is positive already at K=8 (Spearman +0.48 to +0.81 across the six comparisons), confirming that its failure on Qwen at that scale was range restriction, not absence of the gate. The dense cross-family comparison becomes measurable at +0.424 (p = 0.0034); the registered prediction ($\geq$ +0.5) is met in sign and significance and missed in magnitude, and is reported as such. At K=33 the law holds on 528 cells at +0.559 (p = 0.0001) and sharpens to +0.744 on the 19 relations with $W_i \geq 0.6$ on both families. The gradient +0.424 → +0.559 → +0.744 is the law's boundary condition made quantitative: *the* *convergent arrangement is a property of shared knowledge, and it fades, category by category, exactly where knowledge does.* The per-category confirmation is the adherence analysis: at K=33 the correlation between adherence and the representational proxy $\min(W_i^A, W_i^B)$ is Spearman +0.649 (Pearson +0.493), and the three negative-adherence categories sit in or near the unsigned set; at K=8 the same statistic is uninformative and was falsified as registered (eight categories selected for signal compress the proxy's range), a scoping lesson we keep. The proxy is not independent of the quantity it gates (both are attenuated by weak axes), so the behavioral per-relation know-rate was measured on both models (greedy completion, the criterion of `behav_check.py`, 60 prompts per relation). The registered prediction (Spearman $\geq$ +0.4 against adherence) was falsified: the correlation is positive but weak (Spearman +0.21 on the per-relation minimum, +0.26 on the mean, +0.33 for Llama alone), far below the representational proxy's +0.649. We read the divergence as informative rather than fatal: several relations with near-perfect within-category separability verbalize poorly under their single template (P17, P937, P276: within-AUC $\geq$ 0.90, know-rate 0.12–0.42), so the string-match know-rate is a noisy lower bound on what the model represents, and the gate correlates with *represented* knowledge more than with verbalized recall. The sharpest single cell supports the gate at the individual level: P495, the one sign-unstable axis of this study, has the second-lowest behavioral know-rate of all 33 relations (0.017 on Qwen, 0.100 on Llama). Whether the gate is genuinely representational or the know-rate is template-limited is decidable with crossed templates, the same cure already owed to the continent anti-alignment.

Provenance, learned the hard way. During these runs an identity mix-up placed one bundle under another bundle's name. It was caught, and resolved without ambiguity, because a dictionary's cosine matrix is a fingerprint: matrices matched to 5×10^{-4} identify content regardless of filename, and content hashes identify bytes. Two rules join the reproducibility contract as a consequence: every exported artifact carries its full identity (model, K, pairs, seed, revision) inside the file and in the filename, and any bundle whose provenance cannot be established from its content is quarantined from analysis. Names can lie; axes cannot.

What changes for the claims. The arrangement law R5 is unchanged at its registered scope and strengthened beyond it: it now rests on two families, three scales, seven samplings, and one declared gauge. Three amendments are adopted. First, all signed cross-model comparisons in this series use the consensus gauge; the per-category orientation remains correct for within-category use (transfer, AUC) and is retired for cross-model signs. Second, the axis of P495 is documented as sign-unstable (its flip alone poisoned the dense comparison) and the axes of P413 and P30 carry their consensus verdicts, unsigned and semantically negative respectively. Third, the law is stated with its gate: convergence of the truth-mixture arrangement across families is knowledge-gated, and its strength is a monotone function of how well both models know the categories being compared.

23 The measured dictionary and the sparse contract

`dictionary_export.py` saves, per model, the measured dictionary: the per-category oriented axes, the global axis, and the FFN write centroids, with signed matrices and full metadata in one loadable file (a $[K \times d]$ bundle of tens of kilobytes): a K-direction truth and category monitor whose baseline performance is quantified above; the export now spans, per model, the three scales of Sec. 22 together with their consensus-gauge signed matrices. The sparse autoencoder that this series hands to engineering is justified only if, against matched nulls, it (1) aligns with the per-category axes, (2) respects the semantic geography, (3) reproduces the signs, including the inverted loading on continent and the depth reorganization, and (4) beats the measured dictionary on class-signal reconstruction and category decoding. Either outcome is content.

24 Leads, declared as leads

A second readable dimension in the mid-knowledge regime. On Llama-3.2-1B at the peak, a held-out logistic on $(\mathrm{Re}, \mathrm{Im})$ reaches 0.783 vs. 0.690 for the single axis; on Llama-3.2-3B phase scores match or beat the axis in the decay zone. In the Qwen models the second dimension never added more than +0.02. Read through the dimensionality relation, this is the expected geometric signature of the mid-knowledge regime: as knowledge decreases, off-axis components become readable. It refines, and does not contradict, the one-dimensionality claim of (Part I), which was always scoped to known facts. No dedicated null on the margin yet.

Per-head anatomy. The attention contribution is a sum of heads through the output projection, and the fixed-frame instrument applies unchanged to each head's slice. Whether the pro-truth propagation at the peak, and the 3B tug-of-war, are carried by a few identifiable heads or distributed across the pack connects this anatomy to [16].

Framing bias with an open lexical confound. A 2×2 moral/prudential design on Qwen2.5-1.5B: framing does not rotate the axis (transfer 0.979–0.99), but there is a large level bias among equally-true statements (-2.75σ at layer 11), and the SVD of mixed-framing pairs recovers a framing direction at $|\cos| = 0.912$ in PC2, the first part's recovery structure on a new factor. The confound: the bias is already -2.86σ at layer 1, which points to the lexical surface; crossed templates are required before this is more than a lead.

Late-layer decoupling. In all inspected models the total volume written by the FFN grows toward the output while its projection on the truth frame goes to zero and the intact truth AUC decays toward chance. The terminal regime is a decoupling: the model writes enormously, and none of it speaks of truth. Terminal sign changes in the gap are zero-crossings of a dying signal, below both the stability floor and the null, and must not be read as structure.

Frame-aware re-reading of the split. The observational split of Table 11 is measured under a peak-fit frame and should be re-read against pre frames; the causal freezes are frame-free and the attribution stands regardless.

25 Limitations

Four models, two families; a third family would promote the arrangement law from law-candidate to law and widen the relational law's base. The pre frame removes the direct containment circularity but not indirect paths (accumulated earlier writes shape the pre frame). All measurements use last-token states over one dataset (CounterFact, pinned revision) with one statement form per relation; template syntax may contribute to the continent anti-alignment, and crossed templates are the known cure. No permutation nulls were run on the provenance matrices themselves; the diagonal magnitudes sit far above the null scales measured on identical statistics, but the formal check is future work. Ablation measures necessity for linear readability of truth, not for behavior; the freezes act at the reading position only. The category section uses 60 pairs per relation, no bootstrap error bars, and one degenerate relation on Llama (declared). The know-rate is a string-match lower bound, and its per-relation version correlates only weakly with cross-family adherence (Sec. 22), a divergence crossed templates must arbitrate. The consensus gauge is a spectral relaxation: axes with near-zero consensus components are reported unsigned rather than resolved, and a third family is owed to the gauge as much as to the law. The propagate/overwrite reading is a description of measured alignments, not yet a circuit-level mechanism; why attention inherits frames and the FFN opposes them is the open question this paper sharpens rather than answers.

26 Conclusion

Two movements, one instrument, one discipline. The mechanics: attention propagates truth frames it did not write; the FFN opposes the frame of its moment and deposits the material of the next; the genuine post-peak erosion survives every control and is carried, on all four models, by the SwiGLU value stream. The geometry: the single axis is a semantically signed mixture whose arrangement the model does not inherit from the lexical surface but constructs, and constructs the same way across two families, because the arrangement belongs to the knowledge, not to the machine. Stress tests at three scales added the boundary condition: the convergence is knowledge-gated, and the gauge that reads its signs must itself be declared, because the sign of a weakly known axis is not an observable. Between the two movements sits the correction that makes them trustworthy: a pre-registered claim of ours, confirmed on three models, was exposed as the special case of a better law by a control built against our own instrument, and the paper reports the failure with the same care as the findings, because a result that has survived its author's own best attack is worth more than one that has never been attacked. The handoff to the next stage is concrete: a measured dictionary, a four-criterion contract for any sparse autoencoder that claims to do better, and a per-head question with its instrument already built.

Acknowledgments

The author thanks Boaz Nadler for detailed and constructive comments on an earlier version of Part I, which prompted, among many improvements, the formalization of its Method section and Lemma 1; the standards of definition and precision of those comments are applied throughout this manuscript.

27 Reproducibility, Part I

Artifact discipline (before the commands). Every number in this paper is reproduced by one command; every artifact a command produces must be able to prove what it is. The workflow rules, some of them learned the hard way (Sec. 22), are five. (i) Exported files carry their provenance (model, category count, pairs, seed, dataset revision) both in their metadata and in their filename. (ii) Source folders are append-only; analyses run only on a derived archive whose entries are deduplicated and named from their content, with a content hash as the identity of record. (iii) An artifact whose provenance cannot be established from its content is quarantined, not repaired. (iv) Predictions are written to file before runs, and failed predictions are reported with the same care as confirmed ones. (v) No tool prints an automatic verdict: tools print matrices, margins, and distances, and the reading belongs to the researcher.

Every number in this paper is reproduced by one command. `truth_probe.py` runs the controlled experiments (the positive results and falsification experiments 1–4); `canvas.py` renders the visual observations (experiments 5–7); `anatomy.py`, `anatomy_consolidate.py`, `anatomy_expanded.py`, and `ablation.py` produce the anatomy of Sec. 7 and embed the identity, null, and stability safeguards described there. Each result in `truth_probe.py` is an *isolated* subcommand: running one task loads, extracts, and computes only its own data, so a reviewer who wants to verify one figure runs exactly one command. All scripts load the model with safetensors only and load external datasets as Parquet only, never with remote code. A CUDA GPU is optional.

All scripts are archived on Zenodo (DOI 10.5281/zenodo.20938285) and mirrored at `github.com/Francesco-Marhel/TruthProbe`.

Setup.

```
pip install torch transformers datasets
```

Positive results (Tables 1, 2).

```
# Table 1, top row (curated pairs): held-out AUC, ablation,
#                    full-probe baseline, permutation p-value
python truth_probe.py signal --dataset builtin --with-2d --baseline --perm 200

# Table 1, bottom row (CounterFact / TruthfulQA mix)
python truth_probe.py signal --dataset mix --max-pairs 250 \
       --with-2d --baseline --perm 200

# Table 2, top (affirmative -> negated flip; the 2D polarity structure)
python truth_probe.py polarity

# Table 2, bottom (SVD recovers the polarity direction t_P)
python truth_probe.py recovery
```

Know-rate (Secs. 4.2 and Part II) and the CounterFact row of Table 1.

```
python behav_check.py --model Qwen/Qwen2.5-1.5B --n 200 --seed 0

python truth_probe.py signal --dataset counterfact --max-pairs 250 --baseline \
       --perm 200 --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e
```

The same two commands with `-model` set to `Qwen/Qwen2.5-3B`, `meta-llama/Llama-3.2-1B`, or `meta-llama/Llama-3.2-3B` reproduce the corresponding rows of Table 6.

Controlled falsification experiments (Table 3, 1–4).

```
# experiment 1 (PHASE ablation) is the PHASE column of ‘signal‘ above

# experiment 2: global depth signature (fixed axis), curated pairs
python truth_probe.py domino --dataset builtin --perm 100

# experiment 3: dynamic shade on hard data
python truth_probe.py domino --dataset mix --max-pairs 250 --signal-mag --perm 100 \
       --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e \
```

```
        --rev-truthfulqa 741b8276f2d1982aa3d5b832d3ee81ed3b896490

# experiment 4: per-layer native axes + diagnostic ramp
python truth_probe.py domino --dataset mix --max-pairs 250 --per-layer-axis --perm 100 \
        --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e \
        --rev-truthfulqa 741b8276f2d1982aa3d5b832d3ee81ed3b896490
```

Visual observations (Figs. 1–3, experiments 5–7).

```
python canvas.py --dataset counterfact --layer 16 --out counterfact \
        --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e
python canvas.py --dataset truthfulqa --layer 16 --out truthfulqa \
        --rev-truthfulqa 741b8276f2d1982aa3d5b832d3ee81ed3b896490
```

Each `canvas.py` run writes `<out>_coords.png`, `<out>_plane.png`, and `<out>_3d.png`. Run CounterFact and TruthfulQA *separately*; do not pass `-dataset mix` to `canvas.py`, as mixing injects a spurious sentence-format axis.

Anatomy (Tables 4–5).

```
# per-layer attention/FFN readability (identity check embedded)
python anatomy.py --dataset counterfact --max-pairs 250 --perm 100 \
        --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e

# stability of the alternation across 5 seeds
python anatomy_consolidate.py --dataset counterfact --max-pairs 250 --seeds 5 \
        --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e

# the truth axis inside the expanded FFN space (8960-dim)
python anatomy_expanded.py --dataset counterfact --max-pairs 250 --perm 100 \
        --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e

# causal ablation, the three bands of Table 5
python ablation.py --dataset counterfact --max-pairs 250 \
        --band-start 11 --band-end 15 --readout 15 \
        --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e
python ablation.py --dataset counterfact --max-pairs 250 \
        --band-start 16 --band-end 18 --readout 18 \
        --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e
python ablation.py --dataset counterfact --max-pairs 250 \
        --band-start 19 --band-end 26 --readout 27 \
        --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e
```

Anatomy runs use `float32` (Sec. 4.7); about 6 GB of VRAM is enough for Qwen2.5-1.5B.

Data provenance and safety. The curated pairs and the 2×2 polarity set are embedded in `truth_probe.py`. The external datasets are `NeelNanda/counterfact-tracing` (commit `c945b08...`) and `truthful_qa`, *generation* config (commit `741b827...`), loaded as Parquet only. For bit-exact reproduction the dataset commits may be pinned as shown, or a local file passed with `-file-counterfact <path>`. Add `-h` to any subcommand for its options.

Norm-weighting control (Methods of Part I).

```
# repeat with each model and its peak hidden level:
#   Qwen2.5-1.5B->16  Qwen2.5-3B->17  Llama-3.2-1B->8  Llama-3.2-3B->10
python axis_norm_check.py --model Qwen/Qwen2.5-1.5B --layer 16
```

28 Reproducibility, Part II

All tools import `truth_probe.py` and `anatomy.py` from the same repository and embed the identity, null, and stability safeguards described above; models load with safetensors only and `trust_remote_code=False`; the dataset revision is pinned in Sec. 15. Two rules learned inside this project and now part of its method: the model is passed explicitly to every command (tool defaults differ, and one default produced the accidental replication of Sec. 17.2), and no tool prints an automatic verdict. The Llama models require accepting the license on Hugging Face. Representative commands; repeat with `-model` and the landmarks of Table 6 for each model:

```
# behavioral label BEFORE geometry, and the axis/probe gap (Table 1)
python behav_check.py --model meta-llama/Llama-3.2-3B --n 200 --seed 0
python truth_probe.py signal --model meta-llama/Llama-3.2-3B \
       --dataset counterfact --max-pairs 250 --baseline --perm 200 \
       --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e

# the flip, consolidated: 5 seeds + permutation null + rotation check (Table 2)
python flip_consolidate.py --model meta-llama/Llama-3.2-3B \
       --axis-block 9 --flip-layer 10 --scan-start 5 --scan-end 15

# axis provenance: post and pre frames, both components (Tables 3-4)
python axis_provenance.py --model meta-llama/Llama-3.2-3B --peak 9 \
       --scan-start 5 --scan-end 15
python axis_provenance.py --model meta-llama/Llama-3.2-3B --peak 9 \
       --scan-start 5 --scan-end 15 --component attn

# causal ablation with fixed-frame and frozen controls (Table 5)
python ffn_erosion.py ablate --model meta-llama/Llama-3.2-3B \
       --band-start 10 --band-end 12 --readout 12

# exact gate/value split and causal freezes (Table 6)
python swiglu.py attrib --model meta-llama/Llama-3.2-3B \
       --axis-block 9 --scan-start 6 --scan-end 13
python swiglu.py gatefreeze --model meta-llama/Llama-3.2-3B \
       --band-start 10 --band-end 12 --readout 12

# gauge-invariant circuit norms (weights only, CPU; Table 7)
python circuits.py --model meta-llama/Llama-3.2-3B --dtype bfloat16 \
       --peak 9 --flip 10

# category geometry, arrangement law, dictionary export (Sec. 9-10)
python categories.py --model Qwen/Qwen2.5-3B --peak 16 --write-layer 17 \
```

```
--k-relations 8
python categories.py --model meta-llama/Llama-3.2-3B --peak 9 --write-layer 10 --k-relations 8
python arrangement_law.py
python dictionary_export.py --model Qwen/Qwen2.5-3B --peak 16 --write-layer 17 \
--k-relations 8
python dictionary_export.py --model meta-llama/Llama-3.2-3B --peak 9 \
--write-layer 10 --k-relations 8

# dictionary export at any scale (K, pairs per relation, seed)
python crea_dizionario.py --models Qwen/Qwen2.5-3B --k-relations 33 \
       --pairs-per-relation 60 --seed 0 --out-dir dizionari

# consensus sign gauge (Sec. stress; writes <bundle>_gauge.json)
python reorient_gauge.py archivio/*.pt --mode eigen

# arrangement stress test between two gauged dictionaries
python arrangement_stress_test.py --a <A_gauge.json> --b <B_gauge.json>

# behavioral know-rate per relation, both models
python know_rate_per_relation.py --model Qwen/Qwen2.5-3B
python know_rate_per_relation.py --model meta-llama/Llama-3.2-3B
```

`arrangement_law.py` carries the canonical matrices transcribed in its source with the regeneration commands in its header; the transcription was verified cell-by-cell against fresh `categories.py` runs before the statistics reported here.

Scripts are in `src/` of the repository (`github.com/Francesco-Marhel/TruthProbe`); figures `flip_collapse.png`, `provenance_asymmetry.png`, `categories_geometry.png`, `arrangement_law.png` accompany the source.

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